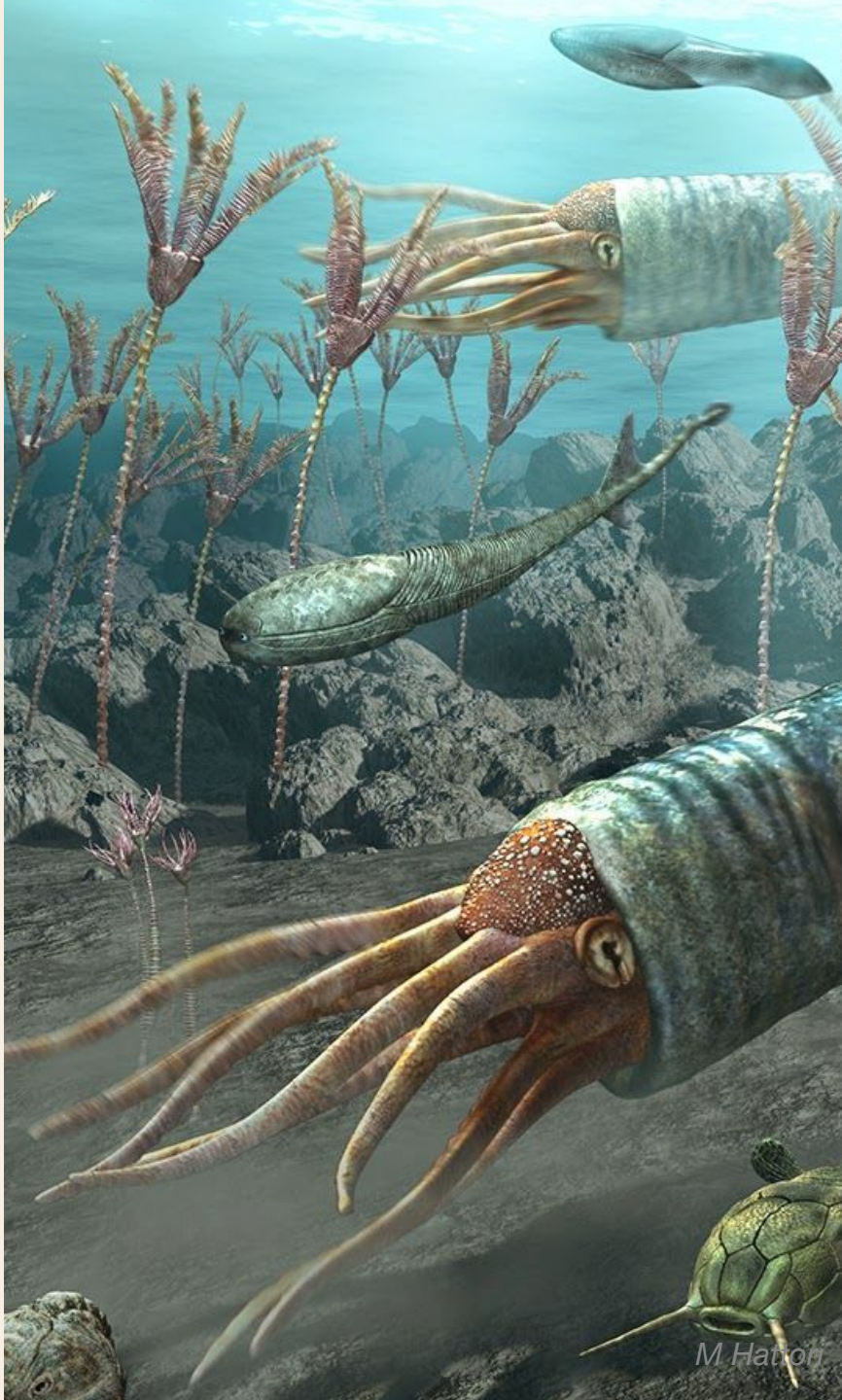
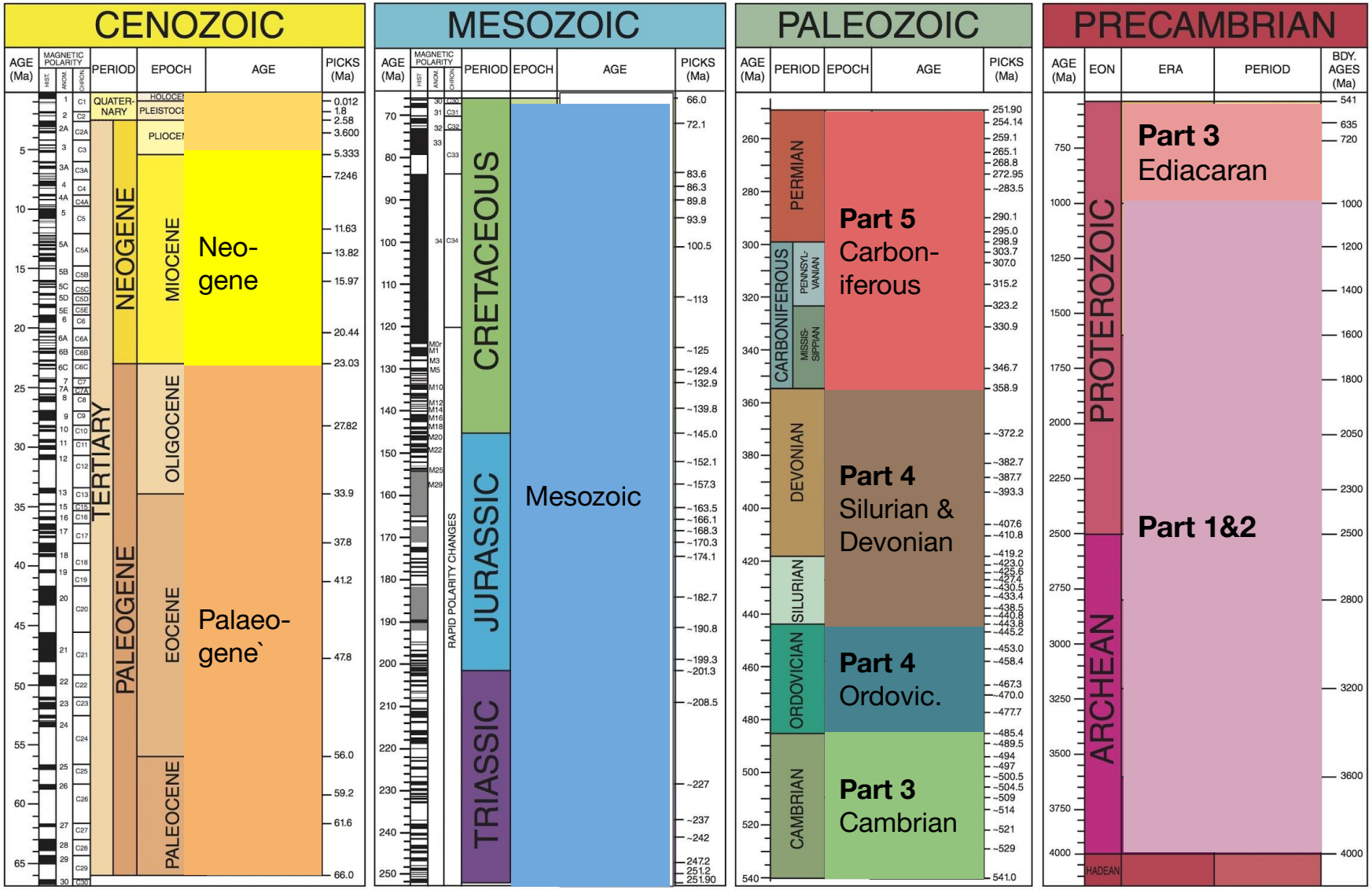
Part 4
Ocean Life

4.1 Reefs



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GSA GEOLOGIC TIME SCALE v. 5.0



Walker, J.D., Geissman, J.W., Bowring, S.A., and Babcock, L.E., compilers, 2018, Geologic Time Scale v. 5.0: Geological Society of America, https://doi.org/10.1130/2018.CTS005R3C.

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*The Pleistocene is divided into four ages, but only two are shown here. What is shown as Calabrian is actually three ages—Calabrian from 1.80 to 0.781 Ma, Middle from 0.781 to 0.126 Ma, and Late from 0.126 to 0.0117 Ma.

The Cenozoic, Mesozoic, and Paleozoic are the Eras of the Phanerozoic Eon. Names of units and age boundaries usually follow the Gradstein et al. (2012), Cohen et al. (2012) , and Cohen et al. (2013, updated) compilations. Numerical age estimates and picks of boundaries usually follow the Cohen et al. (2013, updated) compilation. The numbered epochs and ages of the Cambrian are provisional. A "~" before a numerical age estimate typically indicates an associated error of ±0.4 to over 1.6 Ma.

REFERENCES CITED

Cohen, K.M., Finney, S., and Gibbard, P.L., 2012, International Chronostratigraphic Chart: International Commission on Stratigraphy, www.stratigraphy.org (accessed May 2012). (Chart reproduced for the 34th International Geological Congress, Brisbane, Australia, 5–10 August 2012.)

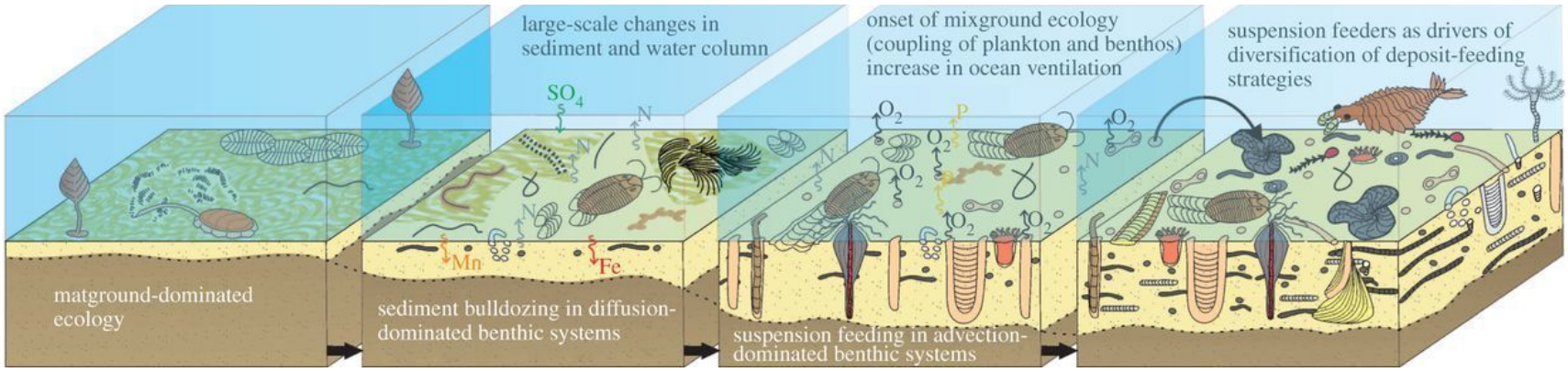
Cohen, K.M., Finney, S.C., Gibbard, P.L., and Fan, J.-X., 2013, The ICS International Chronostratigraphic Chart: Episodes v. 36, no. 3, p. 199–204 (updated 2017, v. 2, http://www.stratigraphy.org/index.php/ics-chart-timescale; accessed May 2018).

Gradstein, F.M, Ogg, J.G., Schmitz, M.D., et al., 2012, The Geologic Time Scale 2012: Boston, USA, Elsevier, https://doi.org/10.1016/B978-0-444-59425-9.00004-4.

Previous versions of the time scale and previously published papers about the time scale and its evolution are posted to http://www.geosociety.org/timescale.

THE GEOLOGICAL SOCIETY OF AMERICA®

Advancing Ecological Structure



Manago et al 2015 Proc B

Ediacaran Ecosystems:
Two dimensional microbial mat surface

Organisms are mostly 'sessile' and live on that flat surface

No burrowing or swimming

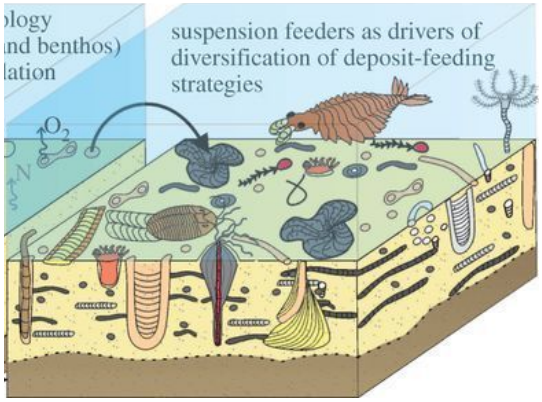
Cambrian Ecosystems:
Burrowing and mixing of layers, penetration of oxygen

Some filter feeders, some swimmers, some predators

More three dimensional

Advancing Ecological Structure

Middle Cambrian



Manago et al 2015 Proc B



Cambrian Ecosystems:
Burrowing and mixing of layers, penetration of oxygen

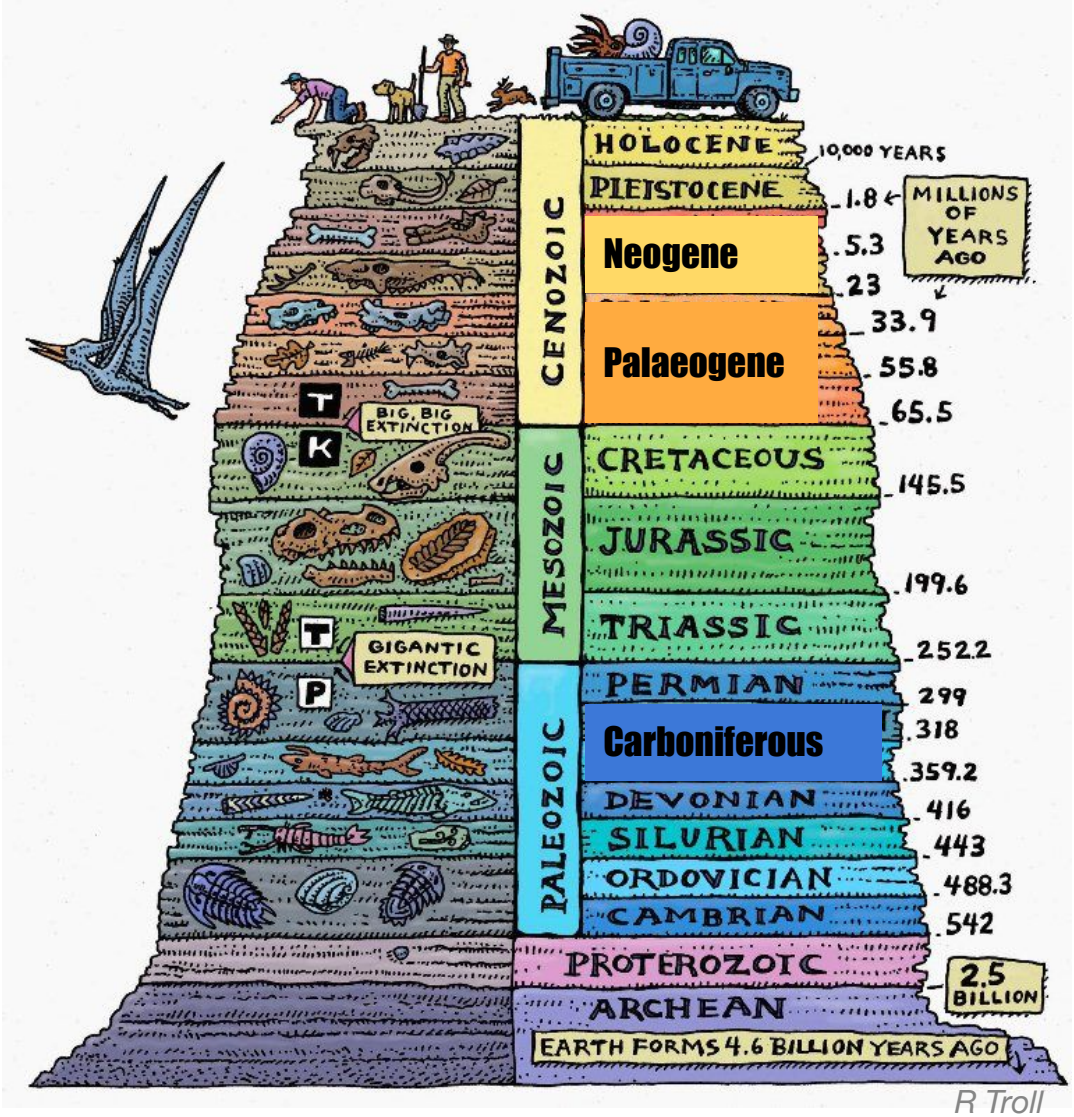
Some filter feeders, some swimmers, some predators

More three dimensional

Modern Marine Ecosystems:
Much more complex

Organisms occupy broader range of lifestyles, diets, and interactions

Geological Time Scale



Eons (millions of years ago):

- Hadean 4600-4000
- Archean 4000-2500
- Proterozoic 2500-540
- **Phanerozoic** 540-today

Ordovician Reefs





Ordovician:
Shift from microbial dominated environments to reef type environments

Corals and sponges dominate creating rocky reef environments:
important geologically and biologically

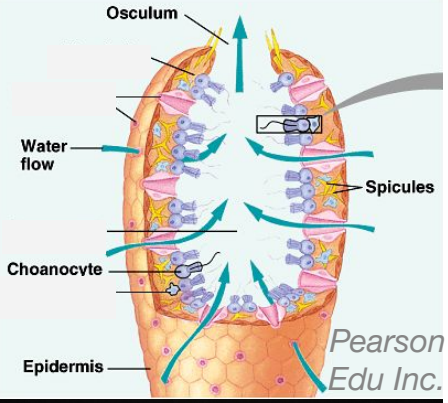
Reef Builders

Sponges are in the phylum Porifera

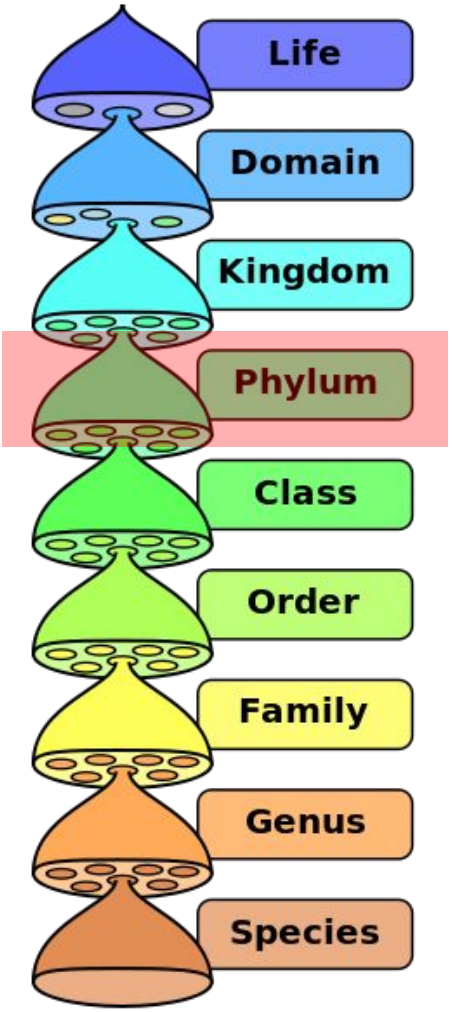
Sponges are very simple animals with simple tissues and no symmetry

Filter feed with ciliary action (Choanocytes)







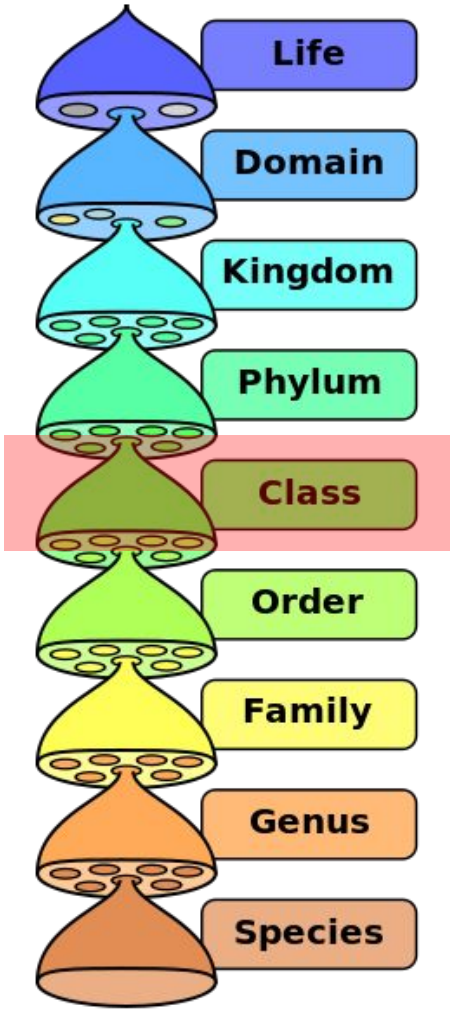


Reef Builders

Corals are *anthozoans:*

Corals are sessile cnidarians that form a *mutualism* with photosynthesising algae (zooxanthellae)

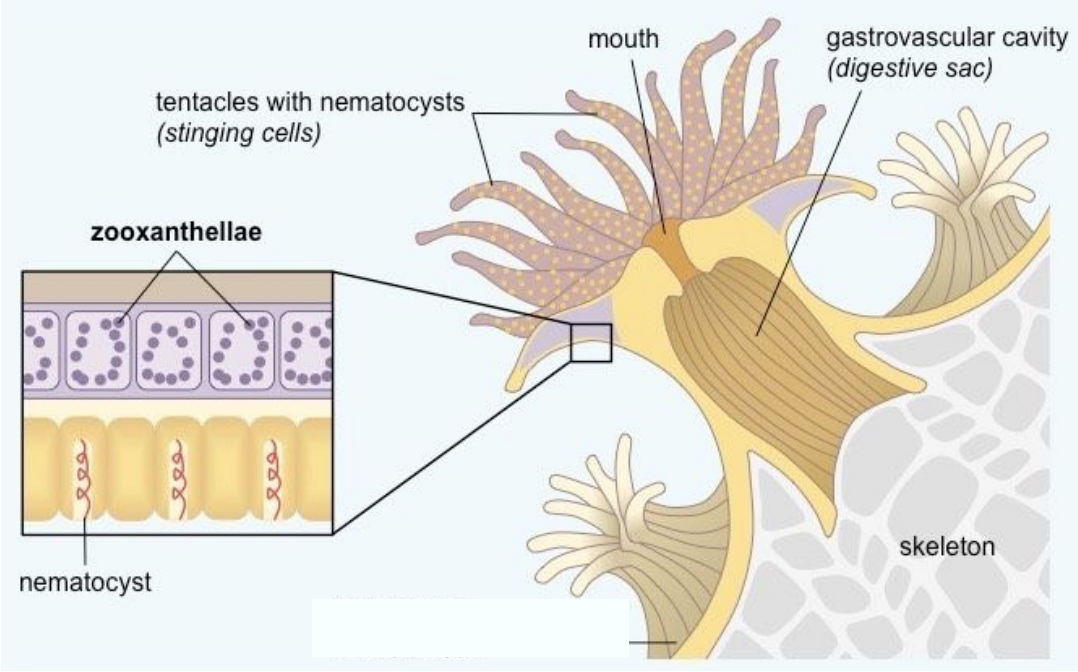




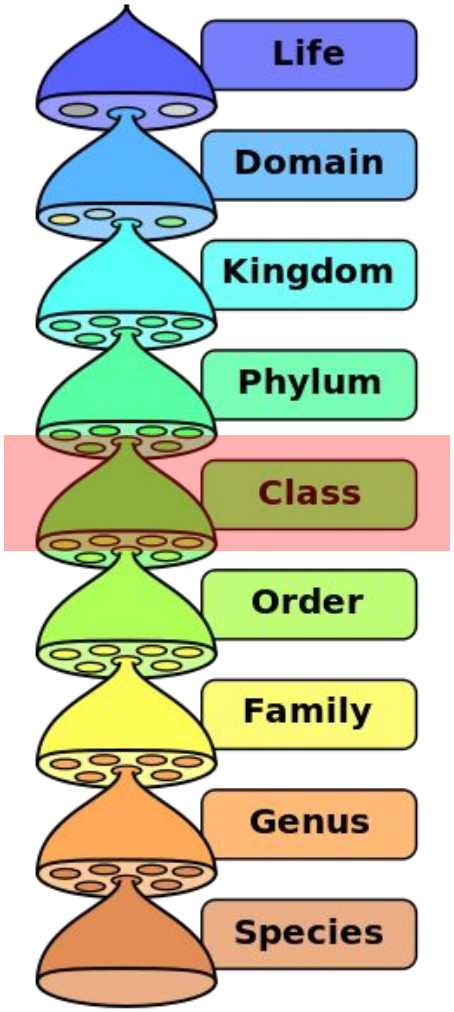
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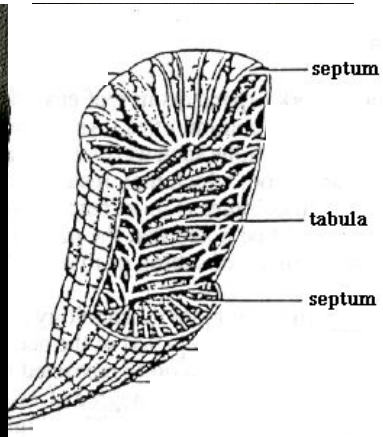
They are comprised of solitary or colonial *polyps*, often housed in or on a mineralized skeleton



Ordovician Reef Builders

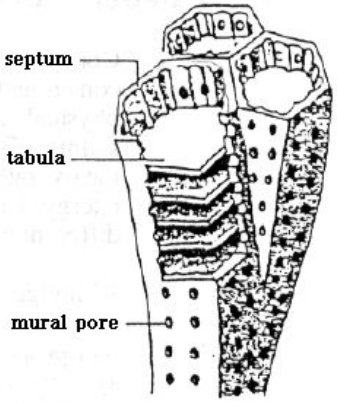
Palaeozoic fossil corals



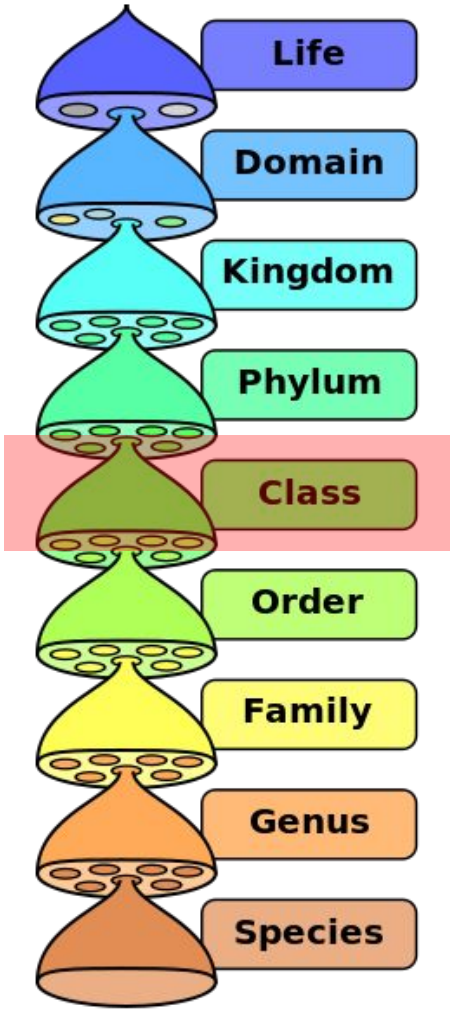


Rugose corals: mostly solitary, known as 'horn corals

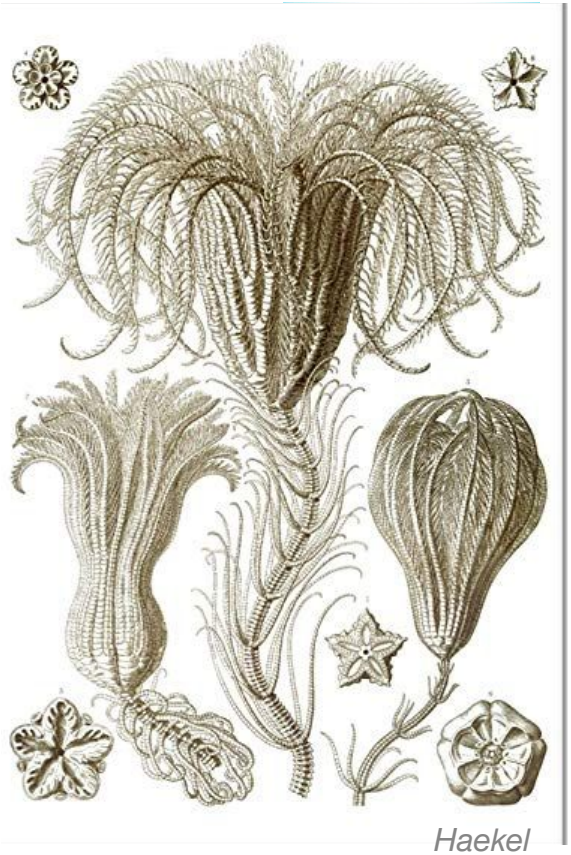




Tabulate corals: colonial with hexagonal cells and partitions



Ordovician Reef Inhabitants

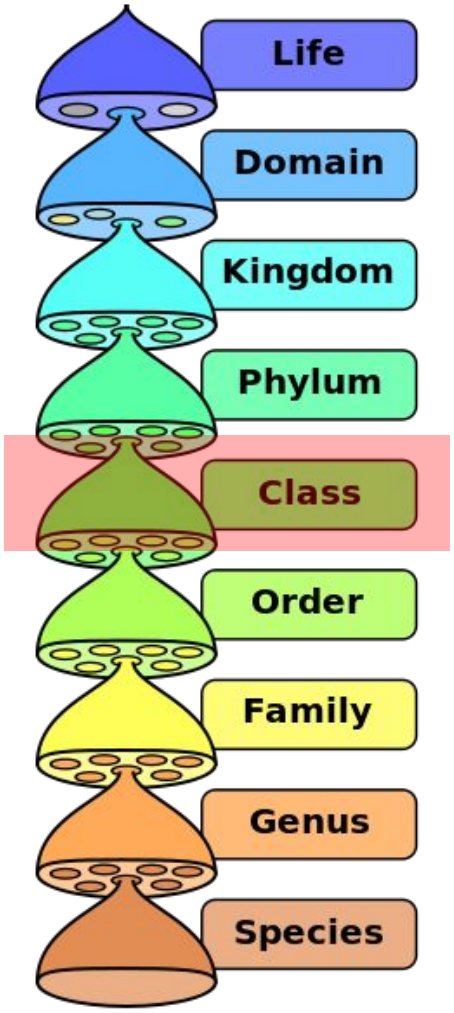


Haekel

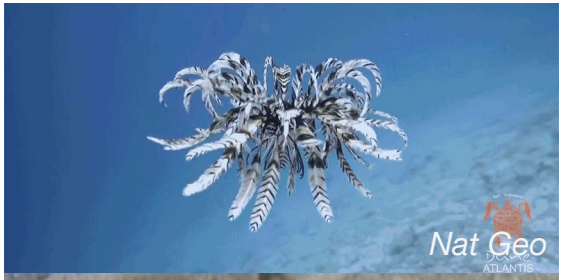
Crinoids (sea lilies): filter feeding echino-derms with stalks



Cephalopods: predatory swimming molluscs, muscular arms



Ordovician Reef Inhabitants







Crinoids (sea lilies): filter feeding echino-derms with stalks

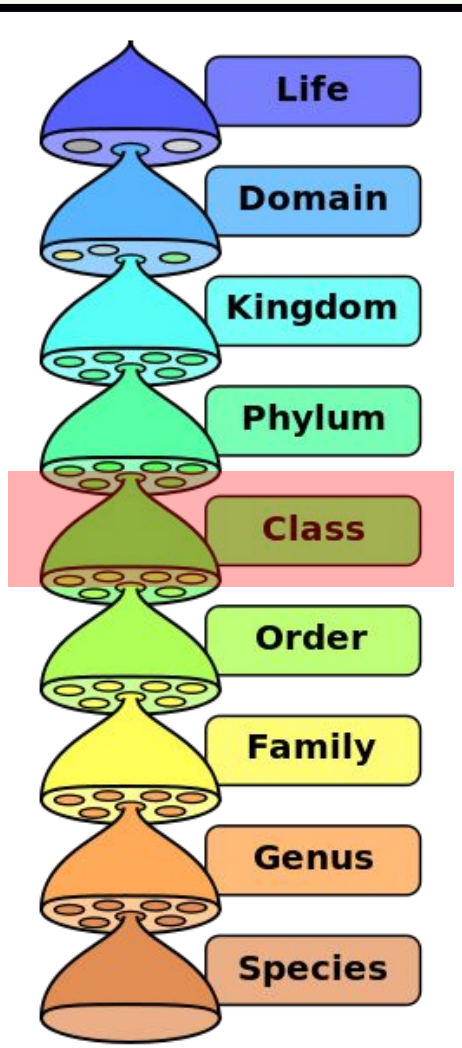




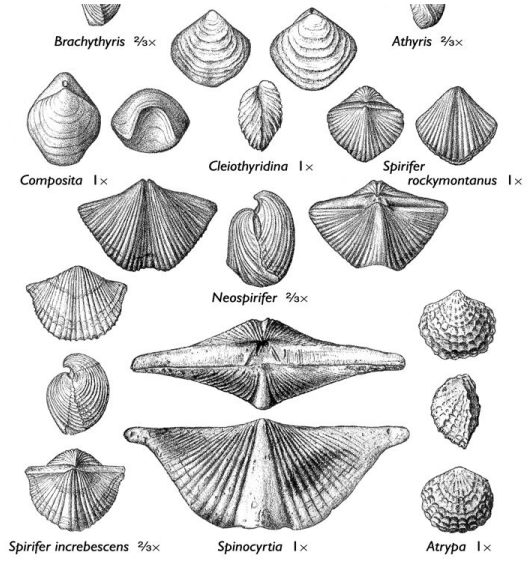




Cephalopods: predatory swimming molluscs, muscular arms

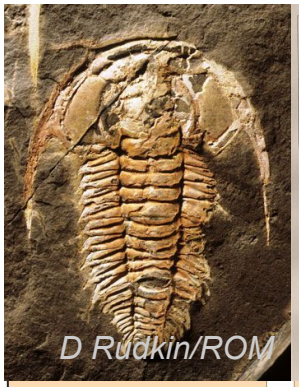


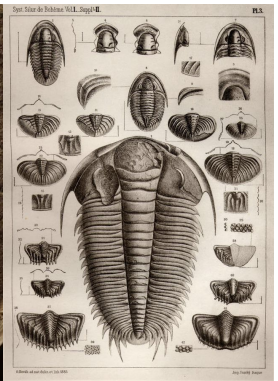
Ordovician Reef Inhabitants

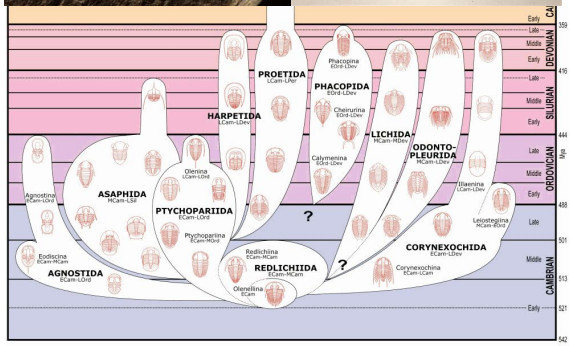




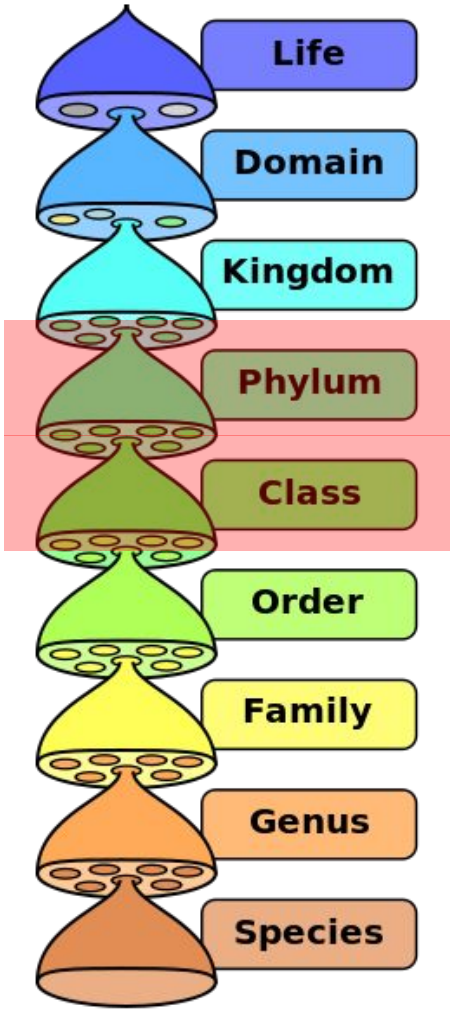
Brachiopods (phylum): sessile filter feeders with front and back valves







Trilobites: arthropods with three-part exoskeleton and varied diets (scavengers, predators etc).



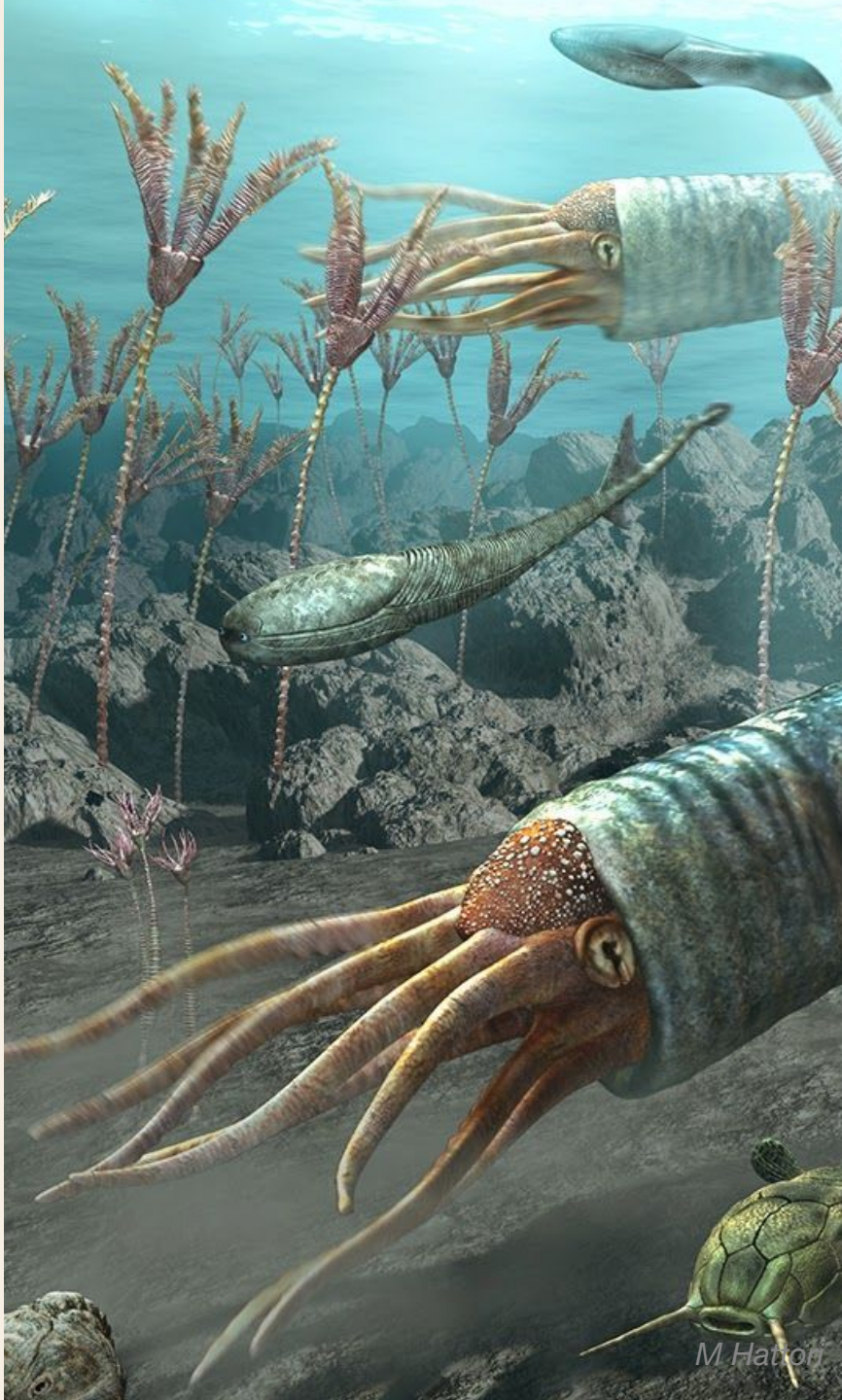

Ordovician Reef Community
Geller-Grimm

Ocean Life
4.1 Reefs

Intended Learning Objective:
Characterise ancient and modern marine reef faunas in terms of major constituent groups and evolutionary changes through time

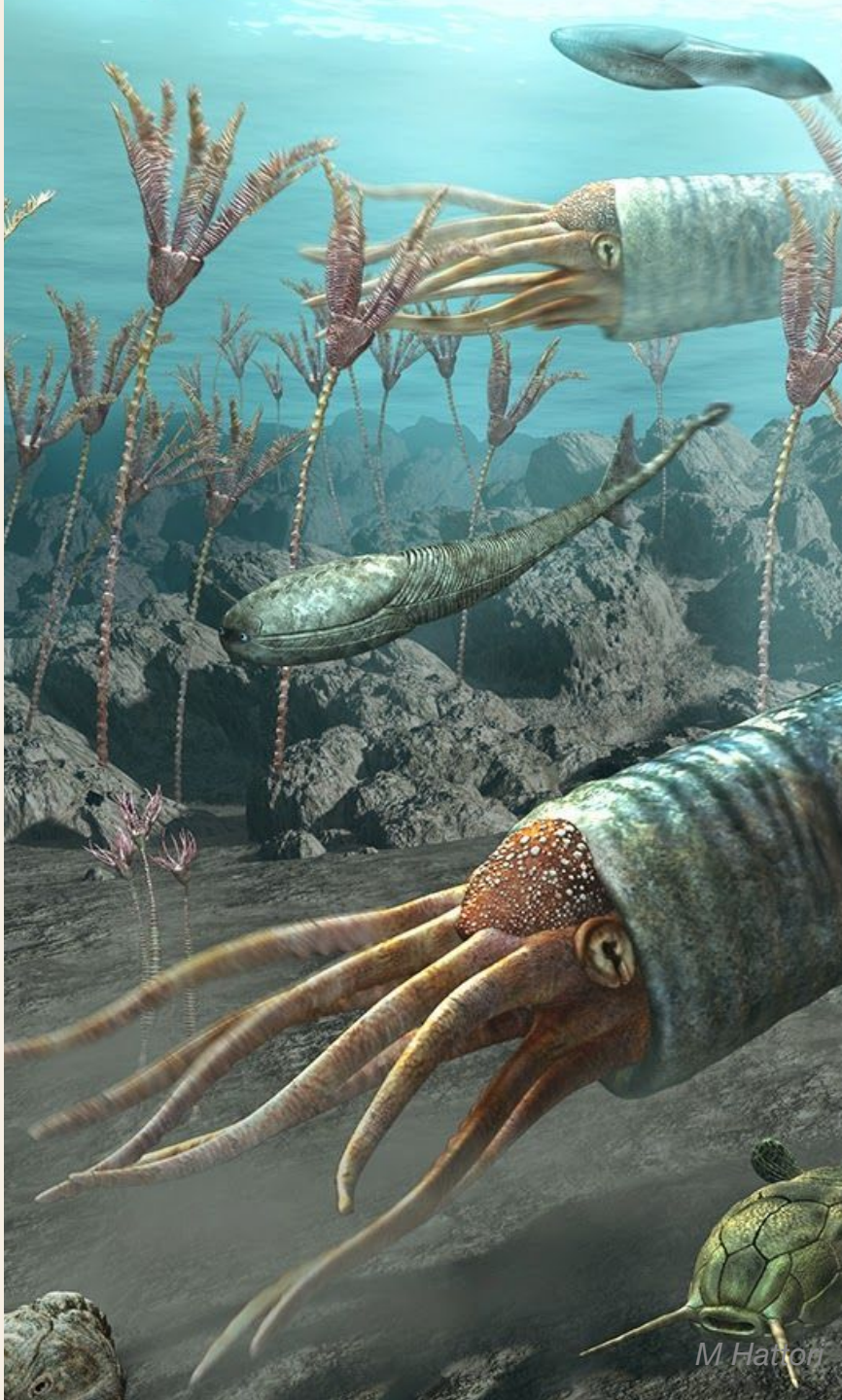
Now take a look at the associated task/question for this section

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Part 3
Ocean Life

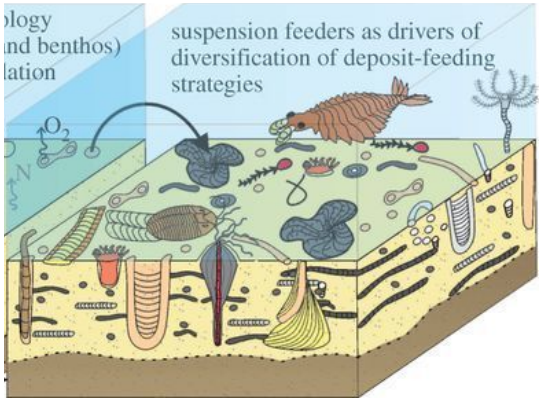
4.2 Ecological Interactions



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Advancing Ecological Structure

Middle Cambrian



Manago et al 2015 Proc B



Cambrian Ecosystems:
Burrowing and mixing of layers, penetration of oxygen

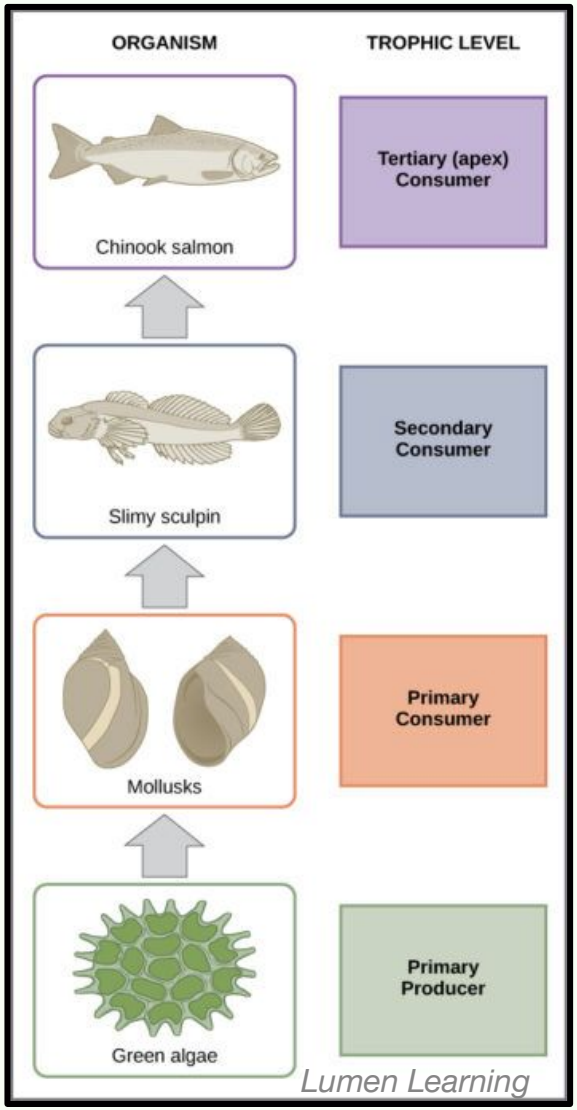
Some filter feeders, some swimmers, some predators

More three dimensional

Modern Marine Ecosystems:
Much more complex

Organisms occupy broader range of lifestyles, diets, and interactions

Ecology and Food webs



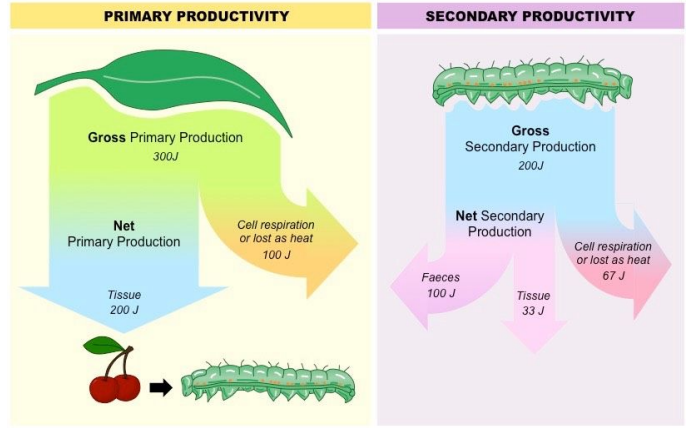
Apex predators only consume carnivores, at the top of the chain

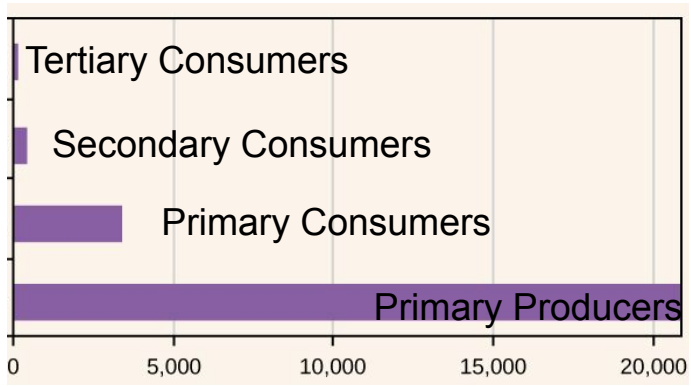
Carnivores consume the herbivores, Omnivores consume primary produces as well as herbivores

Herbivores consume the primary producers (e.g. grazing, filter feeding, etc.)

Autotrophs generate energy and organic matter from sunlight (photosynthesis)

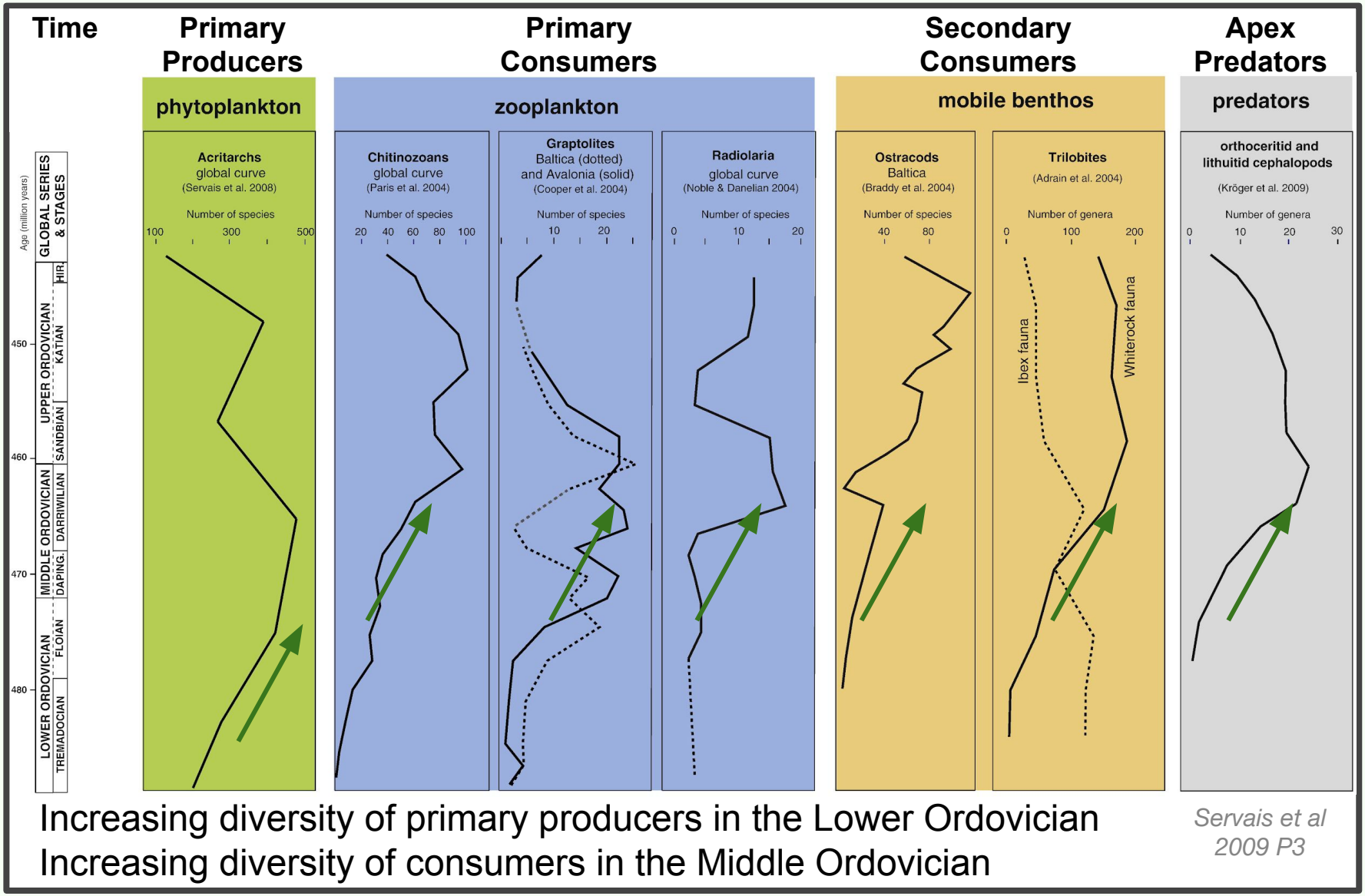
Energy loss at each stage. Most matter is primary producers





Relative Energy Content in Trophic Levels (kcal/m2/year)

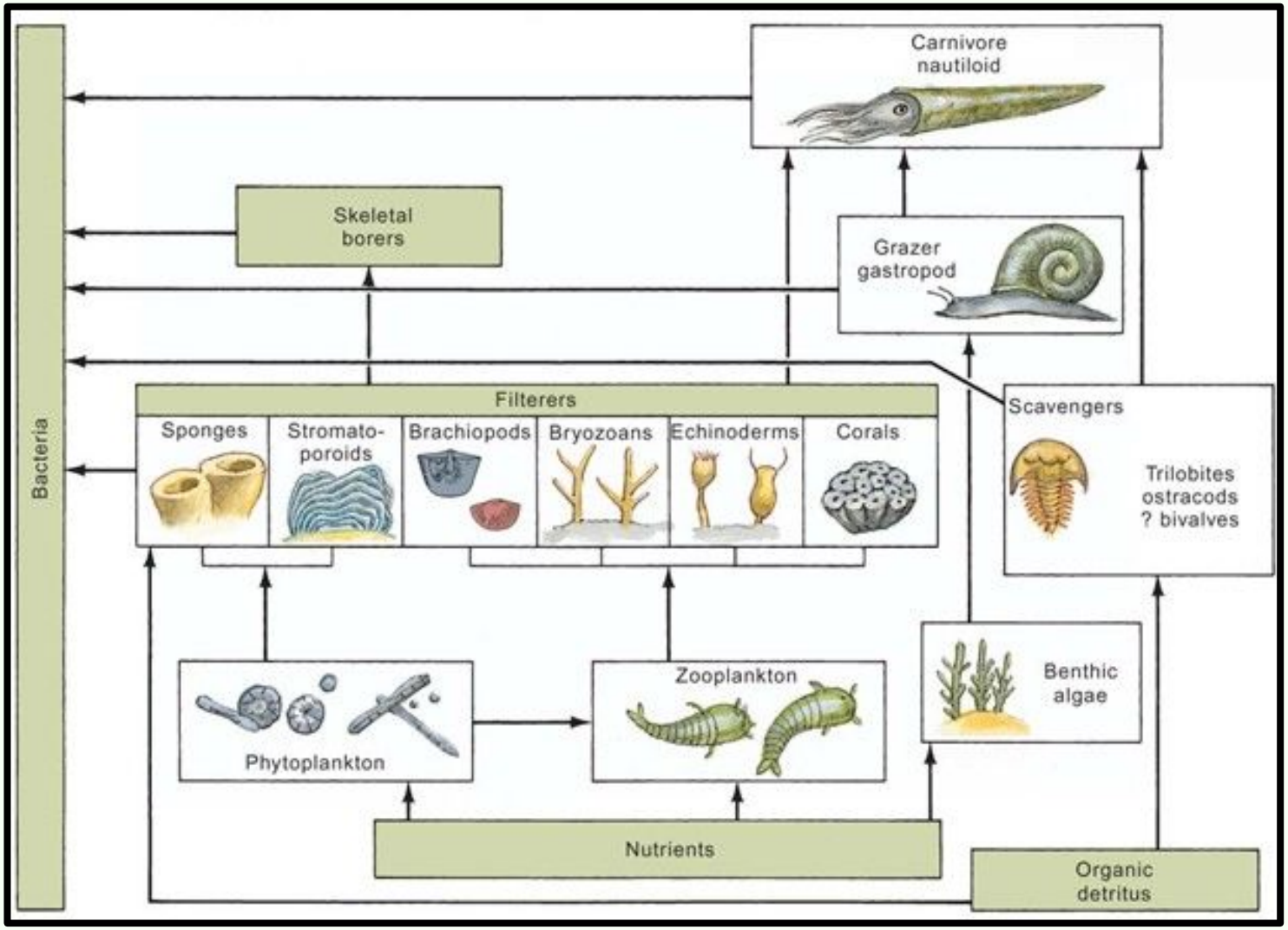
Diversity in Trophic Levels during the Ordovician



Increasing diversity of primary producers in the Lower Ordovician
Increasing diversity of consumers in the Middle Ordovician

Servais et al 2009 P3

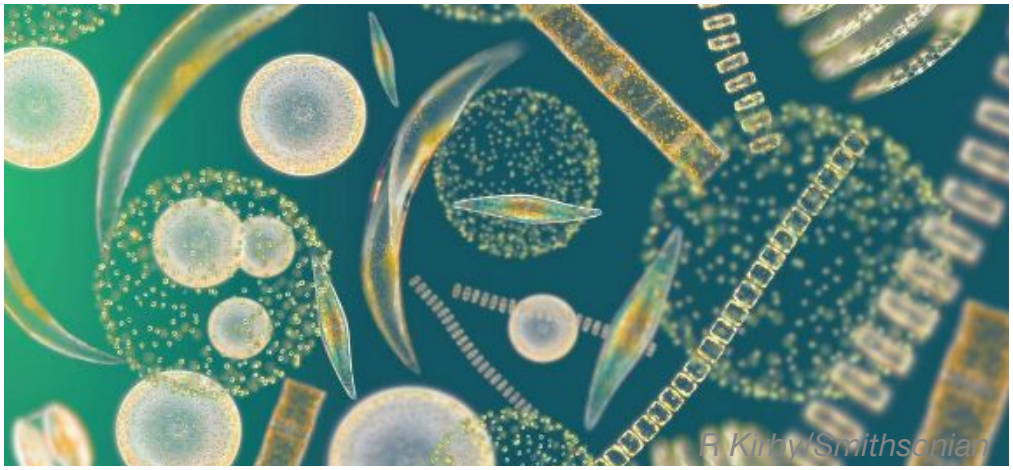
Ordovician Food web



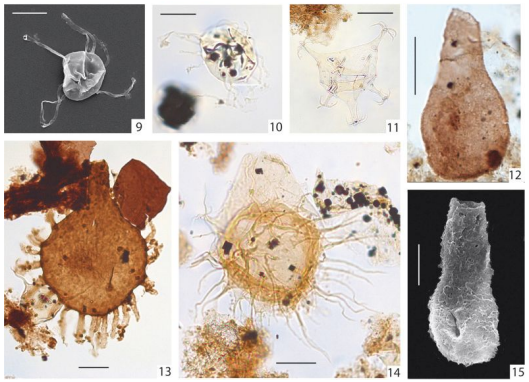
Because the ecosystem is interconnected, increases and diversification of phyto-plankton leads to increasing diversity and complexity throughout the food chain

Plankton

Phytoplankton are autotrophs

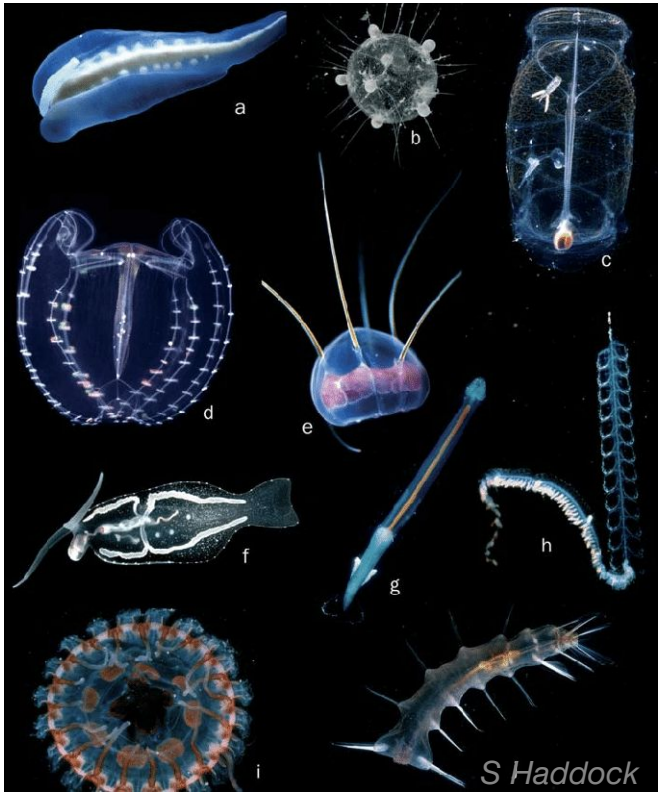


Modern phytoplankton are cyanobacteria and various eukaryotic algae



The Ordovician record of phytoplanton is **acritarchs** - organic walled microfossils of unknown affinity

Zooplankton are heterotrophs



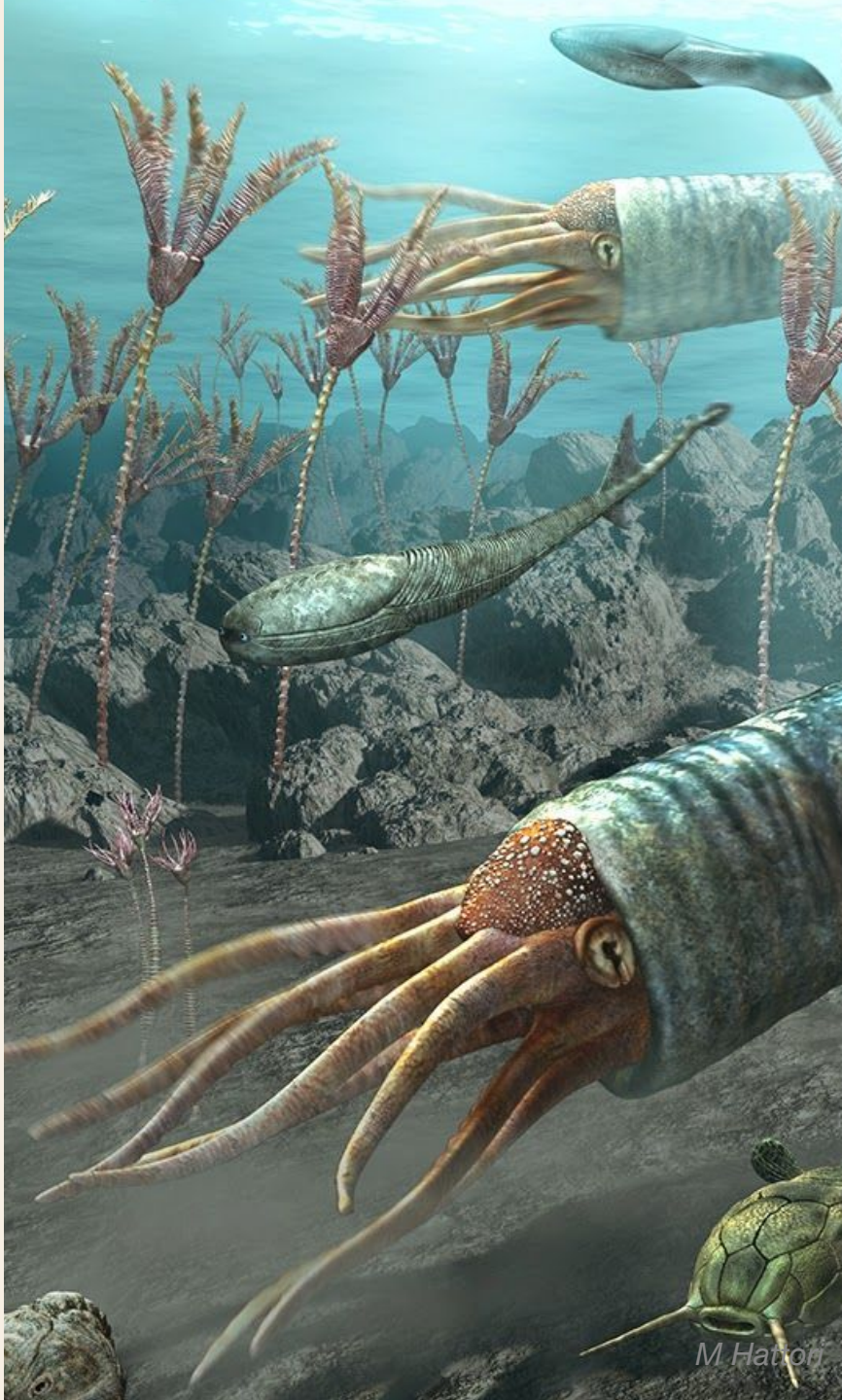
Diverse organisms including tiny arthropods, jellyfish and juvenile molluscs, worms

Ocean Life
4.2 Reefs

Intended Learning Objective: Describe the major roles and interactions of an ecosystem in terms of primary producers and consumers

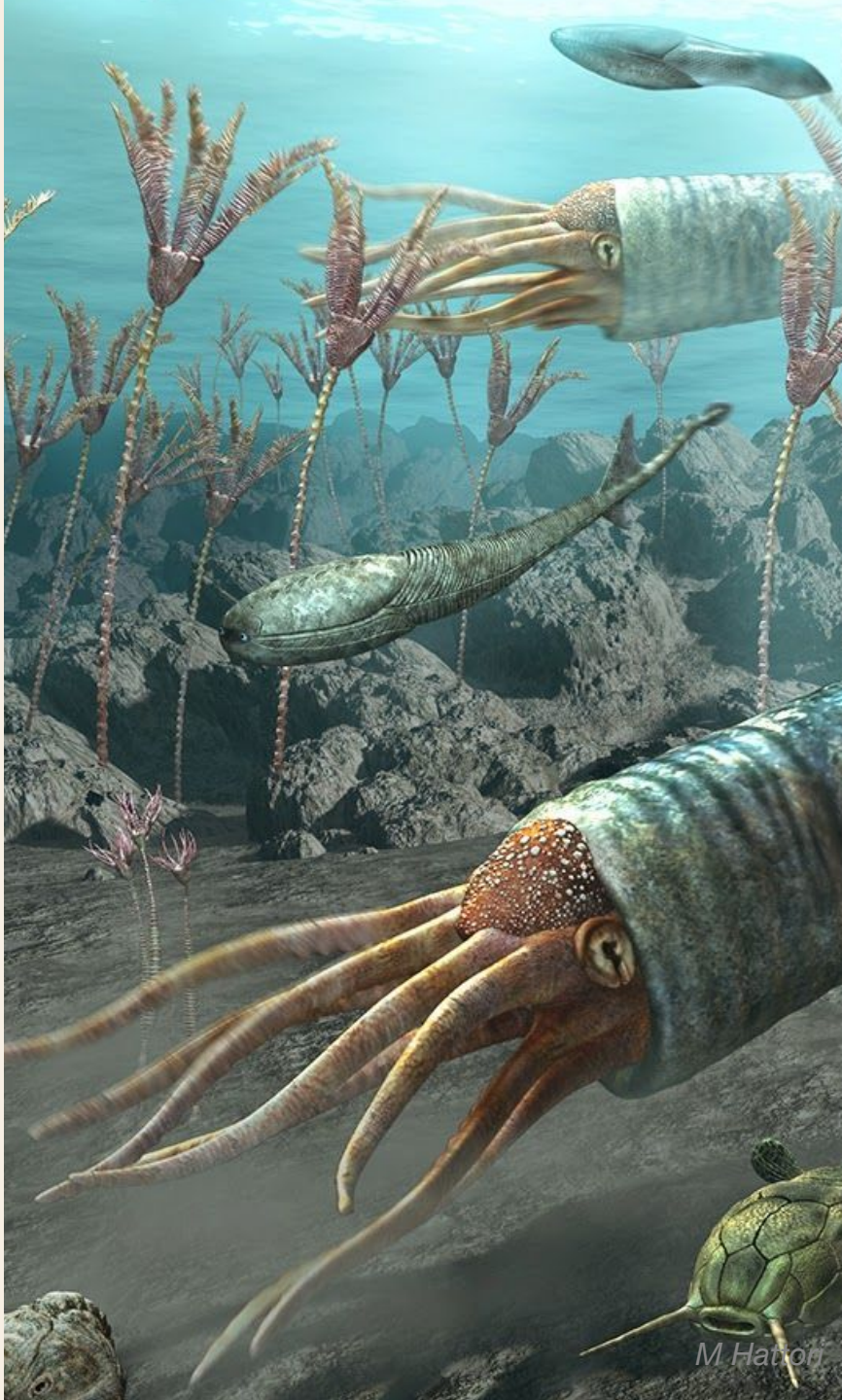
Now take a look at the associated task/question for this section

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Part 4
Ocean Life

4.3 Ordovician Biodiversification



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University of Manchester

Ordovician Reefs



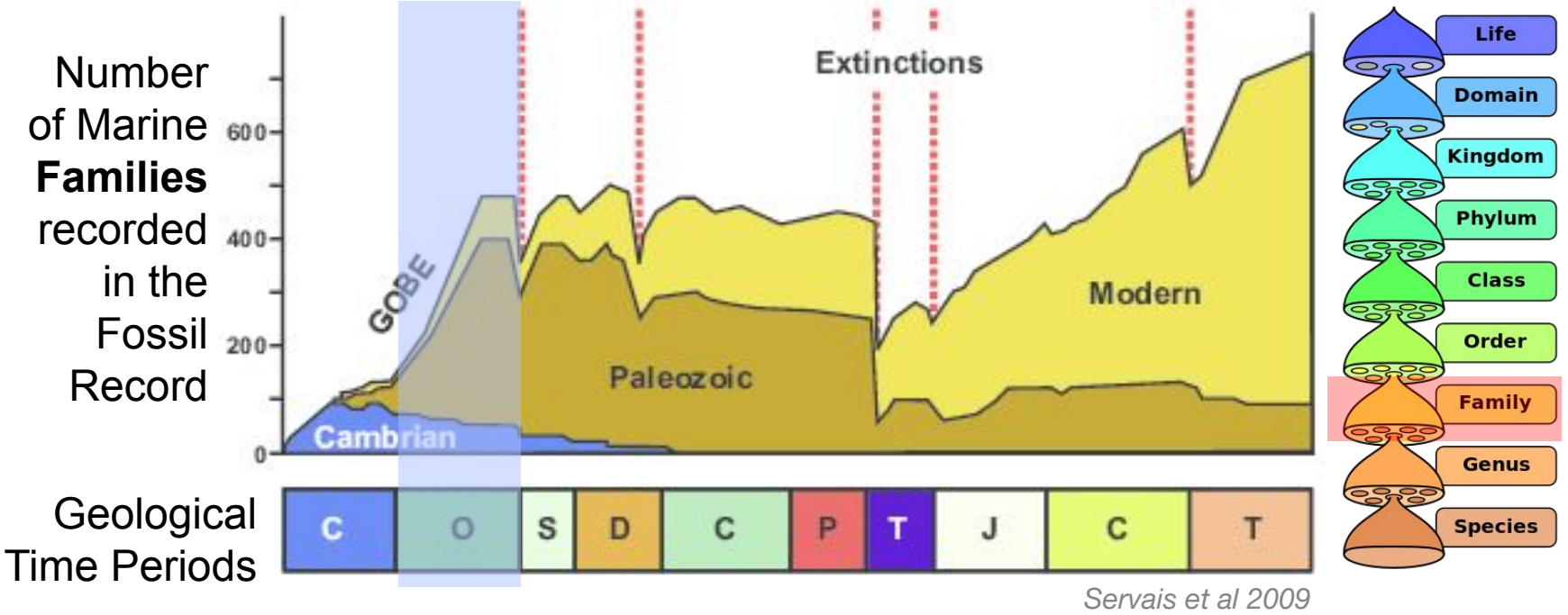


Ordovician:
Shift from microbial dominated environments to reef type environments

Corals and sponges dominate creating rocky reef environments:
important geologically and biologically

Great Ordovician Biodiversification Event

Patterns of biodiversity through time



Servais et al 2009

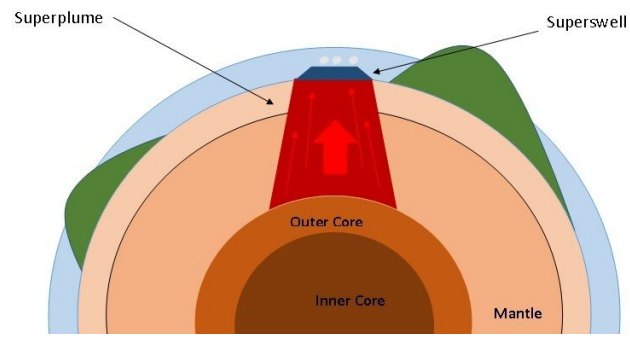
Compilation of fossil data for Phanerozoic shows pattern of originations and major extinctions- **Sepkoski's Curve**

Big increase in number of animal families occurred in the Paleozoic: the **Great Ordovician Biodiversification Event**

Ordovician Earth



Servais et al 2010 P3

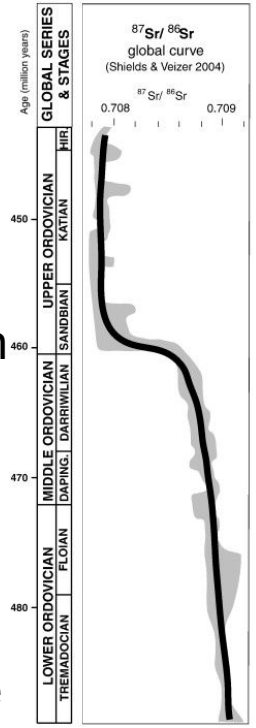


Splitting of the supercontinents & massive **volcanic activity** (mantle superplume)

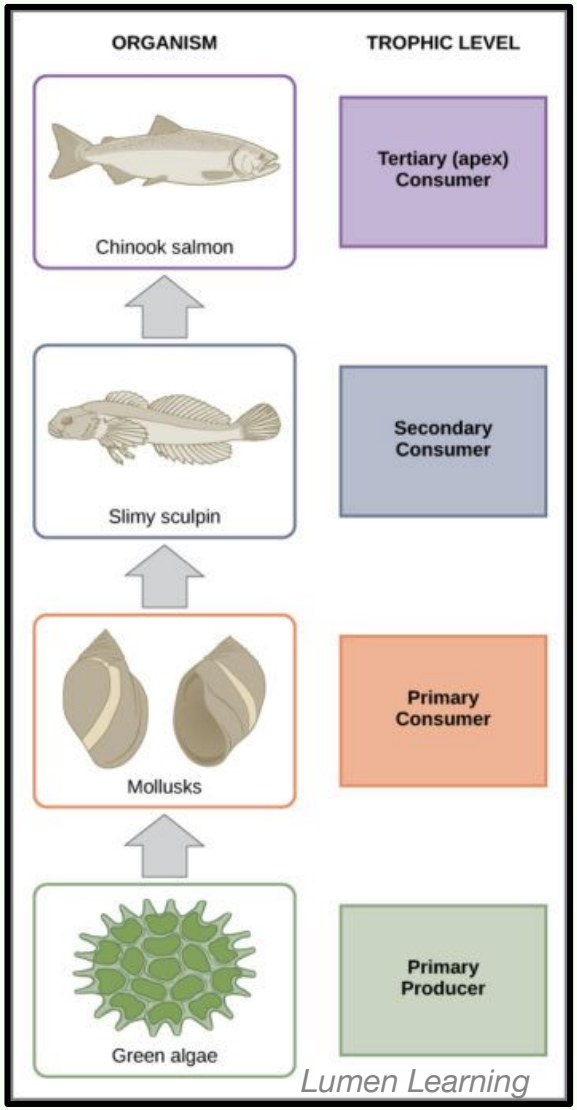
Global warming, high sea levels, and large shelf area

Strontium excursion

All of this is conducive to phytoplankton radiation and thus global biodiversification



Ecology and Food webs



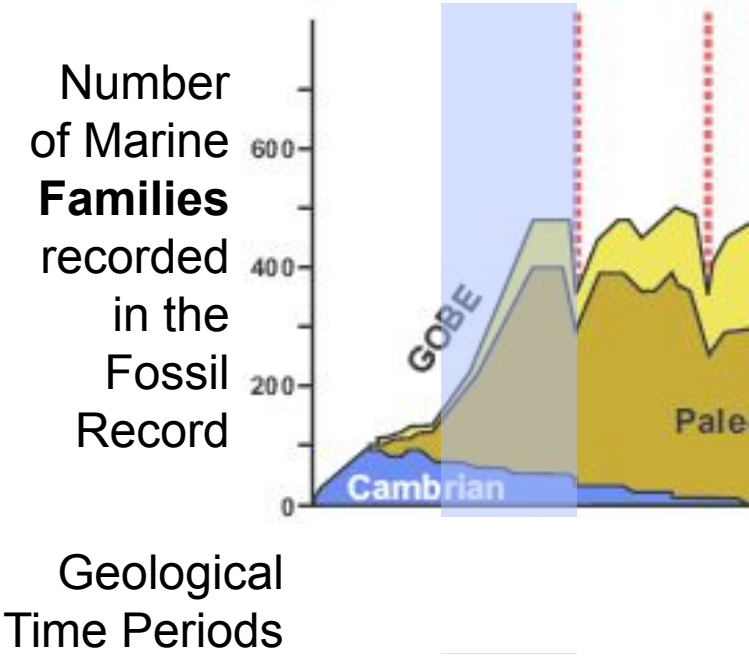
Apex predators only consume carnivores, at the top of the chain

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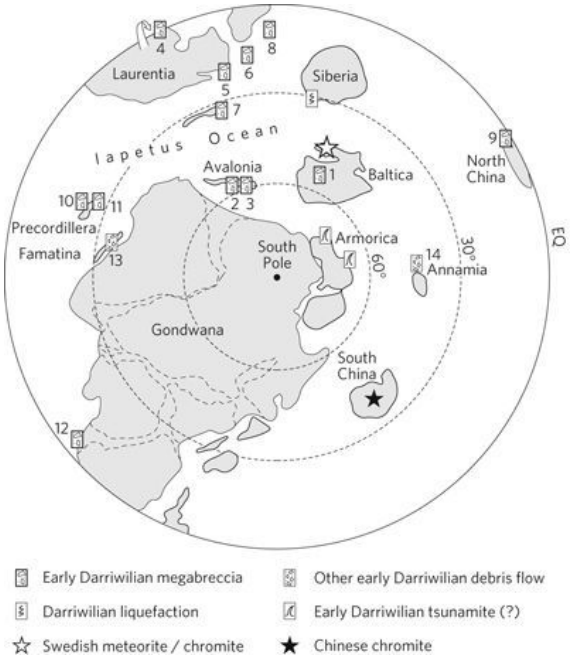
Autotrophs generate energy and organic matter from sunlight (photosynthesis)

Patterns of biodiversity

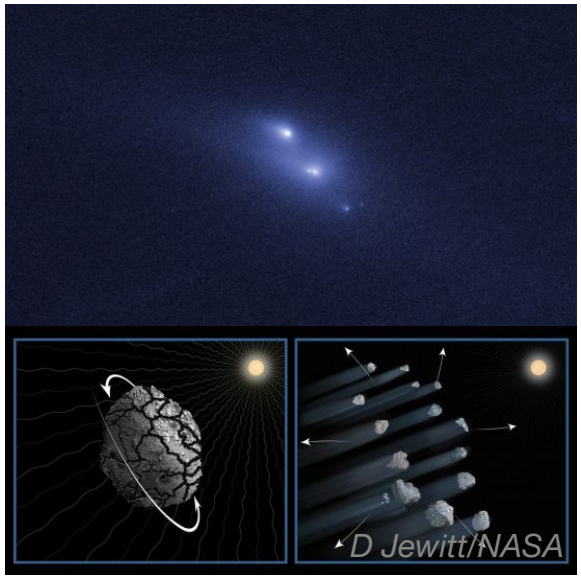


Extraterrestrial Input?

Global record of megabreccias (broken fragments of rock) and chondrites (meteorites) in the Middle Ordovician

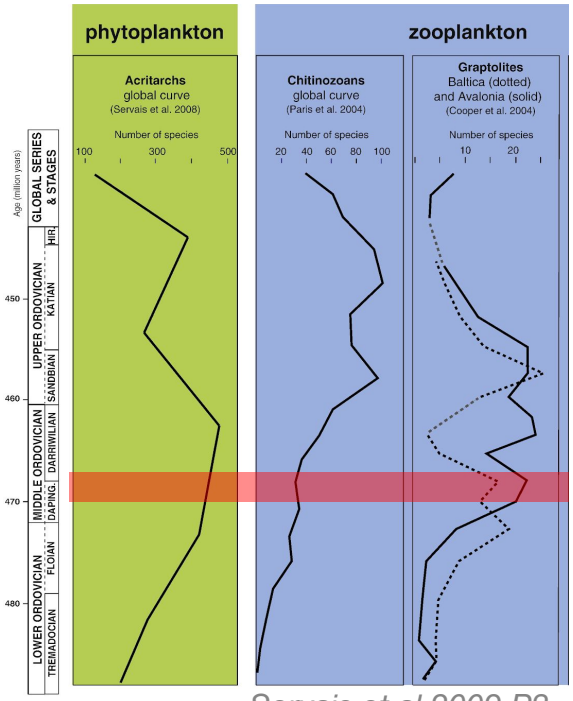


Parnell 2009 Nature Geoscience



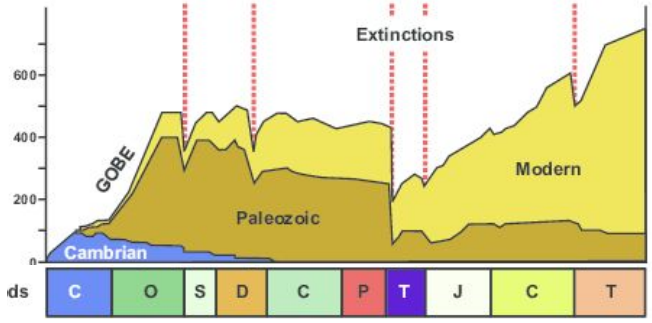
Postulated to be due to break up of a massive asteroid and then a period of heavy meteorite bombardment on earth during that time

The timing of the bombardment relative to the biodiversification and the possible link is not very clear



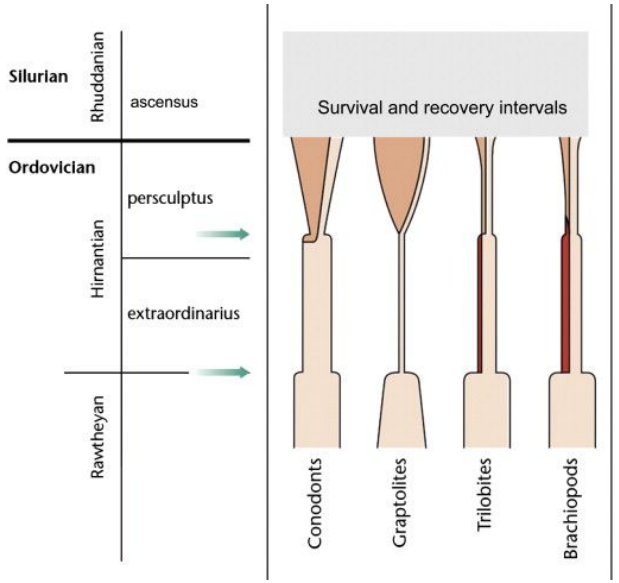
Servais et al 2009 P3

End Ordovician Extinction Event



At the end of the Ordovician was the first of the ‘big five’ mass extinction events

It was actually two pulses of extinction close to each other in time



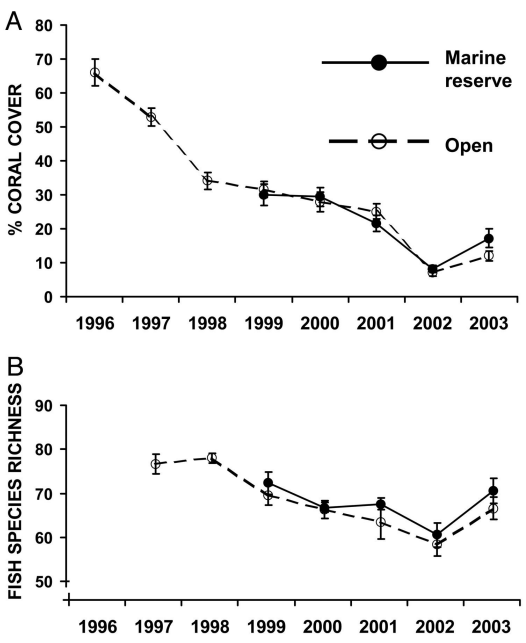


The first pulse was related to global cooling, massive glaciation and drop in sea-level

The second was related to changing ocean cycles and widespread anoxia, warming again

Harper et al 2014 Gondwana Research

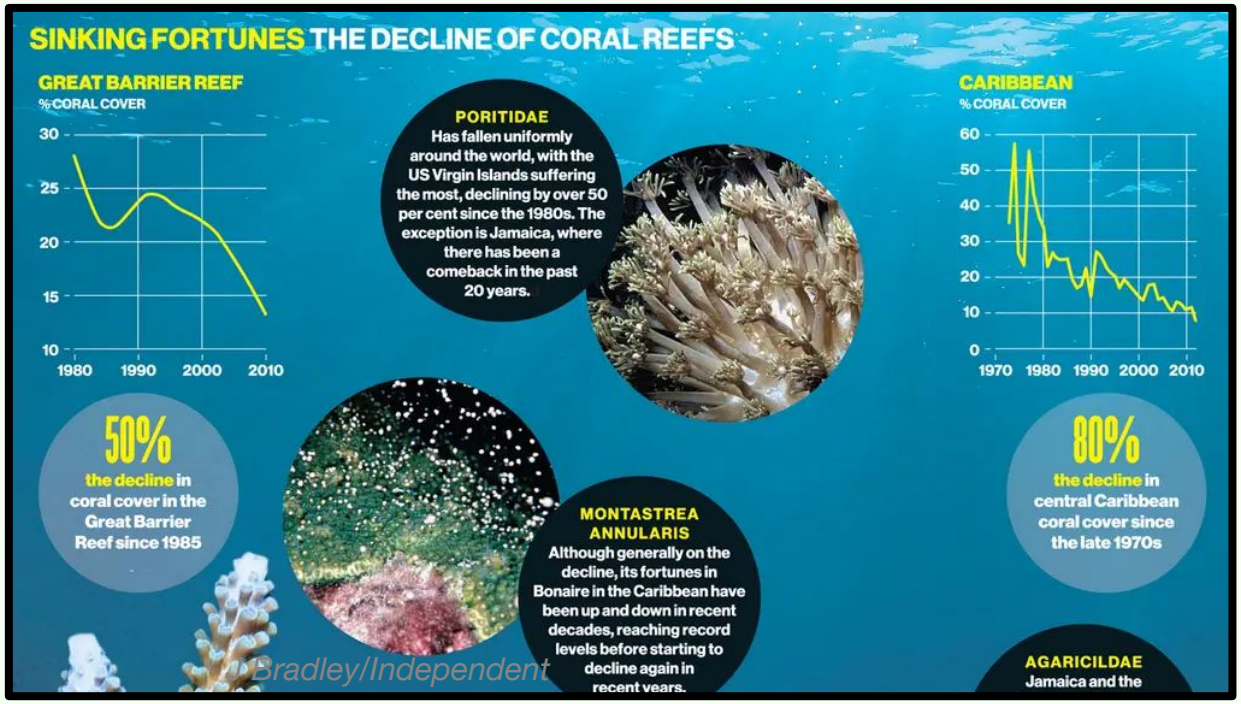
Modern Coral Reefs are in Crisis



Jones et al 2004 PNAS

Modern coral reefs are in decline due to warming, anoxia and ocean acidification

The future?



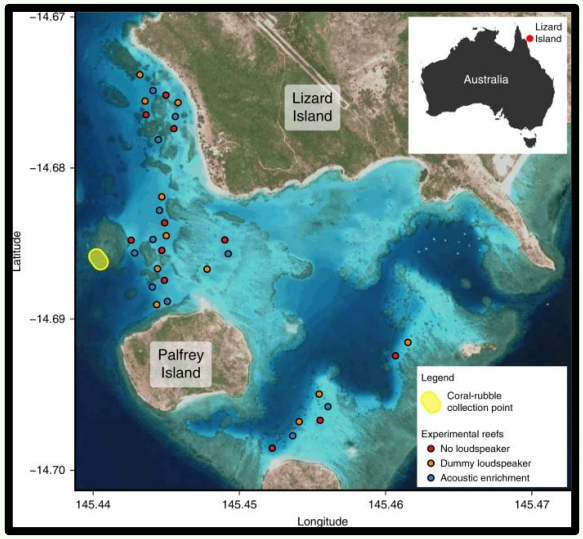
Bradley/Independent

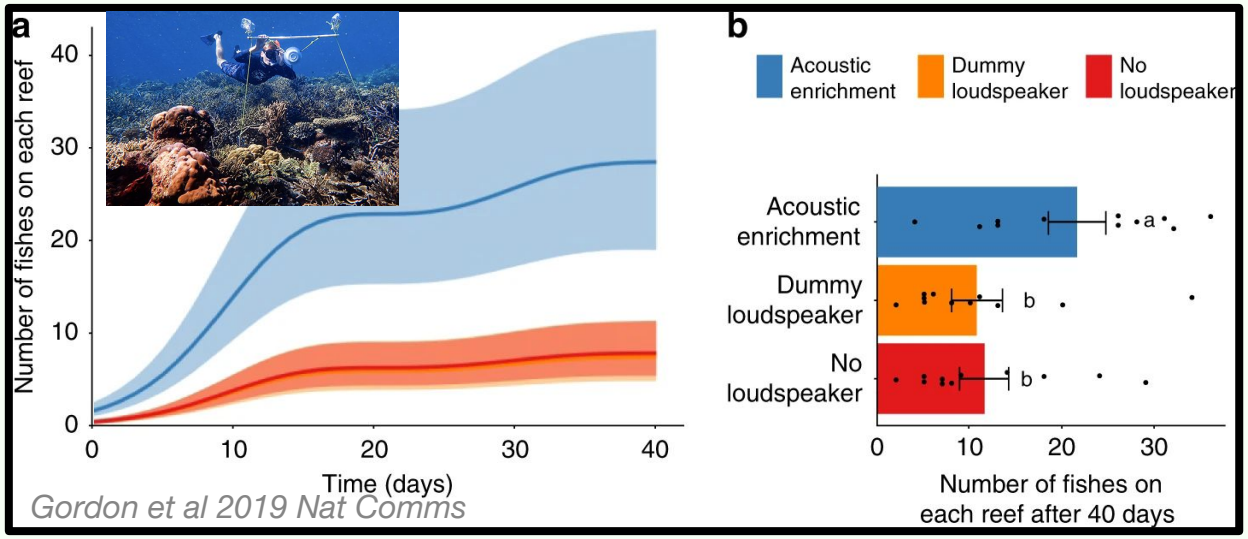


When temperatures rise, **coral bleaching** occurs whereby the coral expel their zooxanthellae algae



An Odd Sounding Experiment



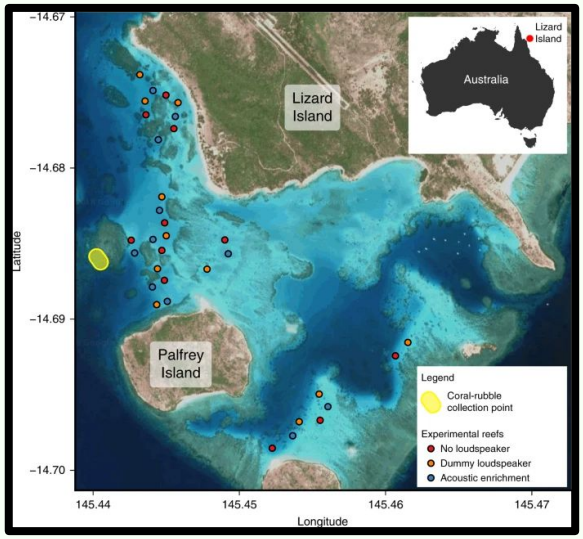


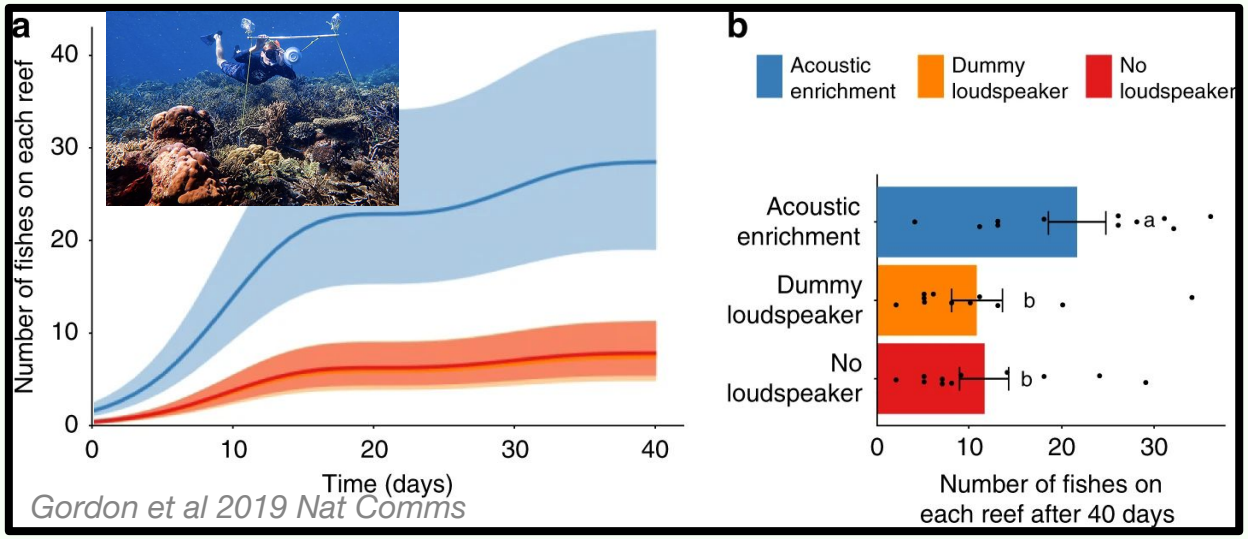
Gordon et al 2019 Nat Comms

An Odd Sounding Experiment

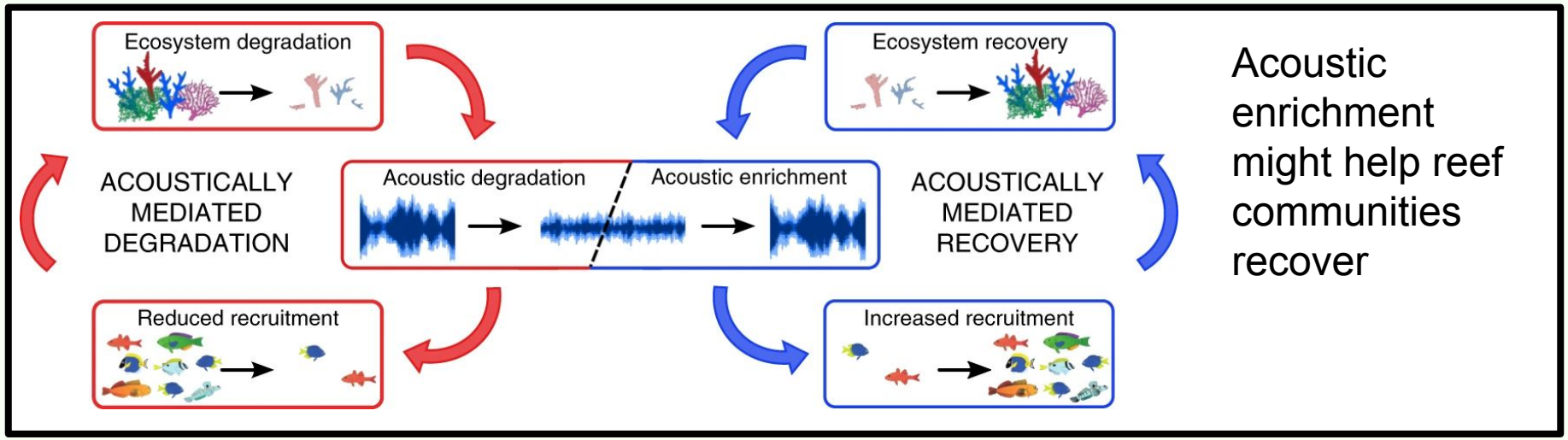


An Odd Sounding Experiment





Gordon et al 2019 Nat Comms



Acoustic enrichment might help reef communities recover

Ordovician Summary

- Modern coral reef type systems first appear in the Ordovician
- The Great Ordovician Biodiversification event saw radiation of brachiopods, trilobites, cephalopods etc in response to radiation of phytoplankton and zooplankton
- Periods of warming and cooling related to radiations and extinctions

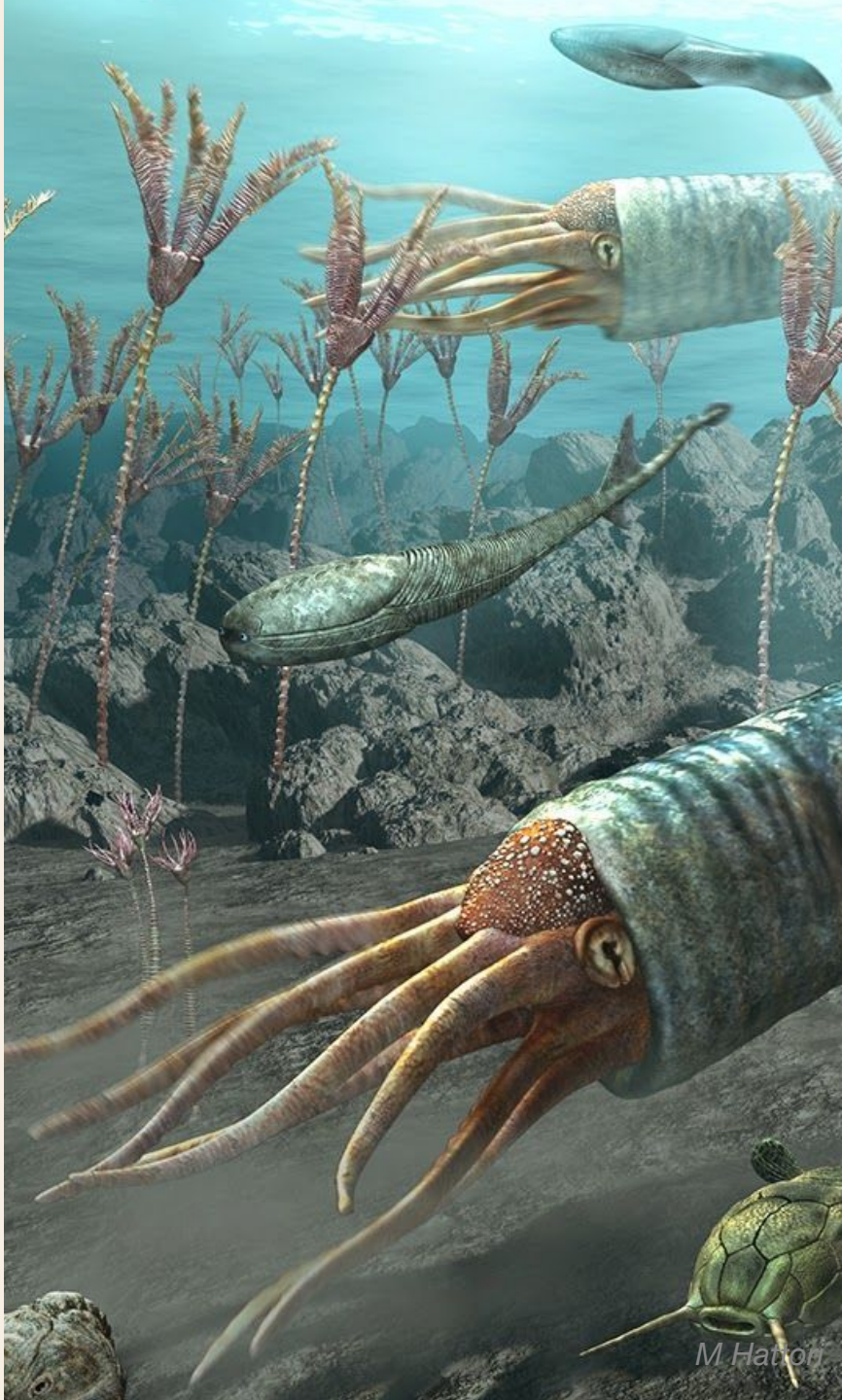


Ocean Life
4.3 Ordovician Biodiversification

Intended Learning Objective: Explain the role of plankton and geological events in the context of the Great Ordovician Biodiversification event and end Ordovician extinction

Now take a look at the associated task/question for this section

Robert Sansom, University of Manchester



Part 4
Ocean Life

4.4 Vertebrates

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University of Manchester



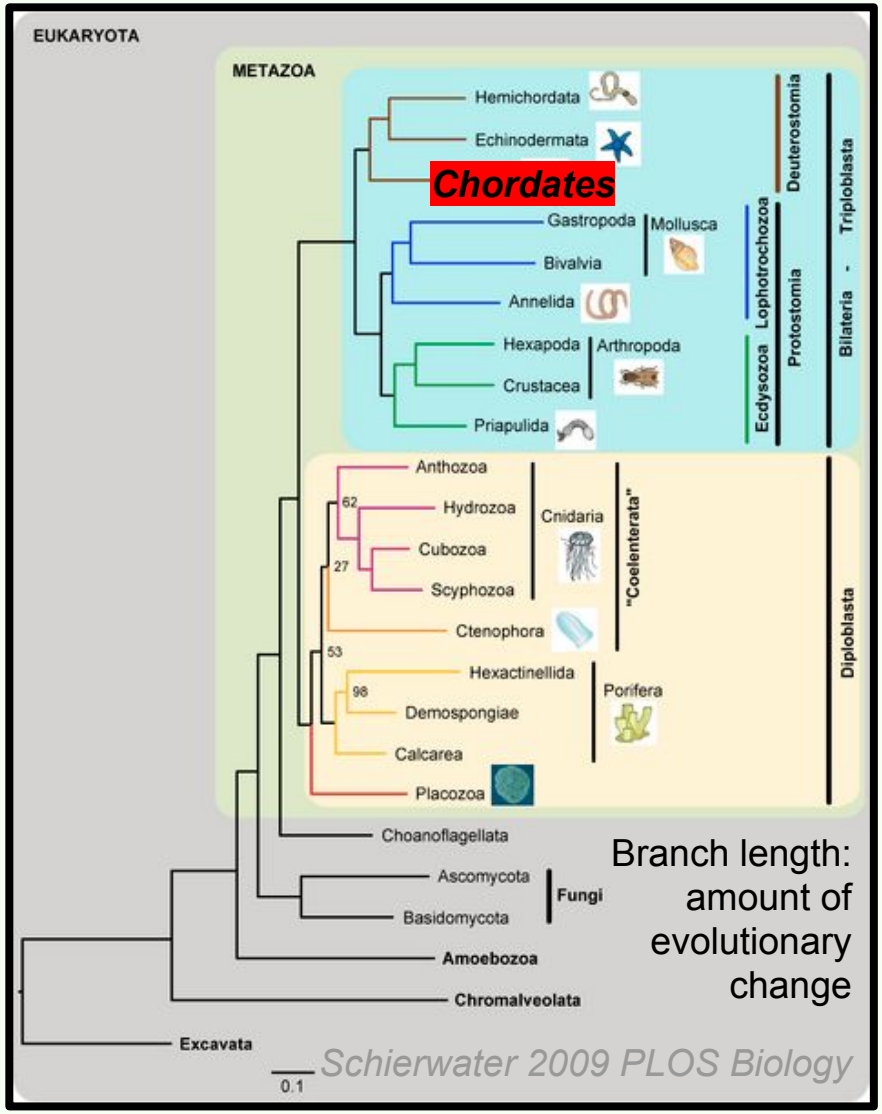
The Vertebrates

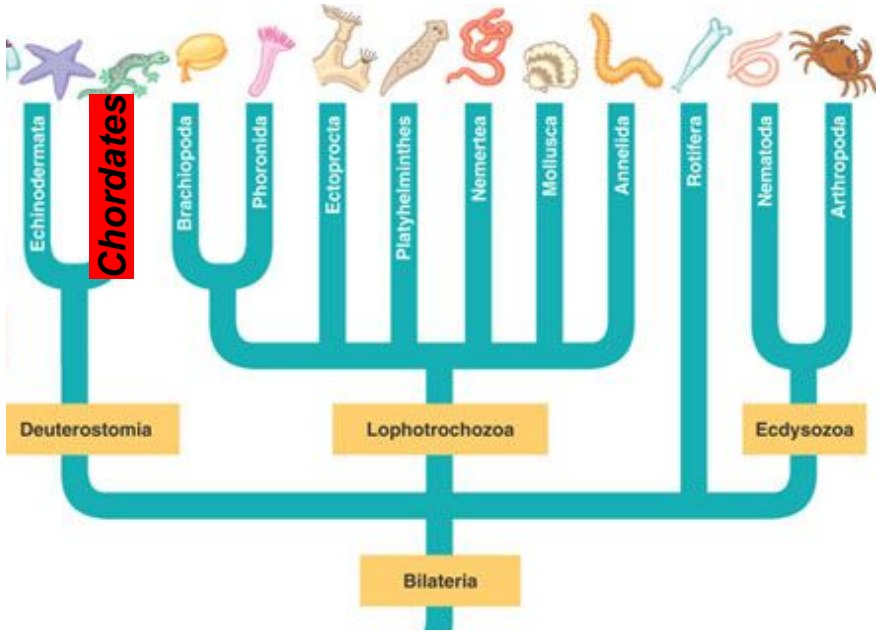
Who are they (*we*)?

Where and when did they (*we*) evolve?



Metazoan Phylogeny

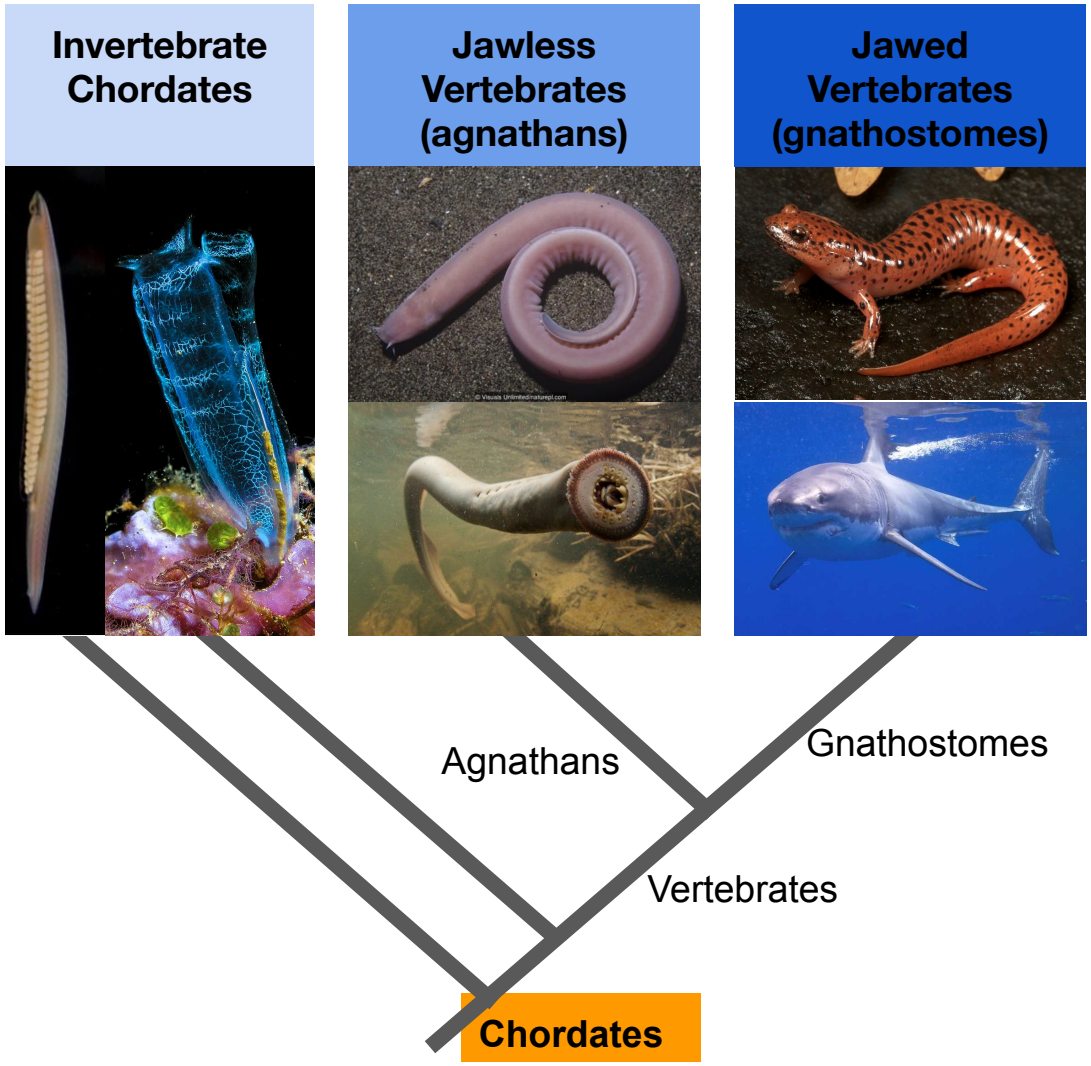




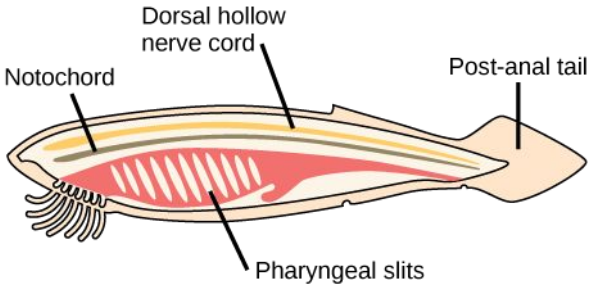
There are three **superphyla** of bilaterian animals (groupings of phyla):

- **Lophotrochozoa** (molluscs, annelids and lots of other phyla)
- **Ecdysozoa** (arthropods & allied phyla)
- **Deuterostomes** (chordates inc. vertebrates, and echinoderms)

Vertebrates



Vertebrates are in the phylum **Chordata**

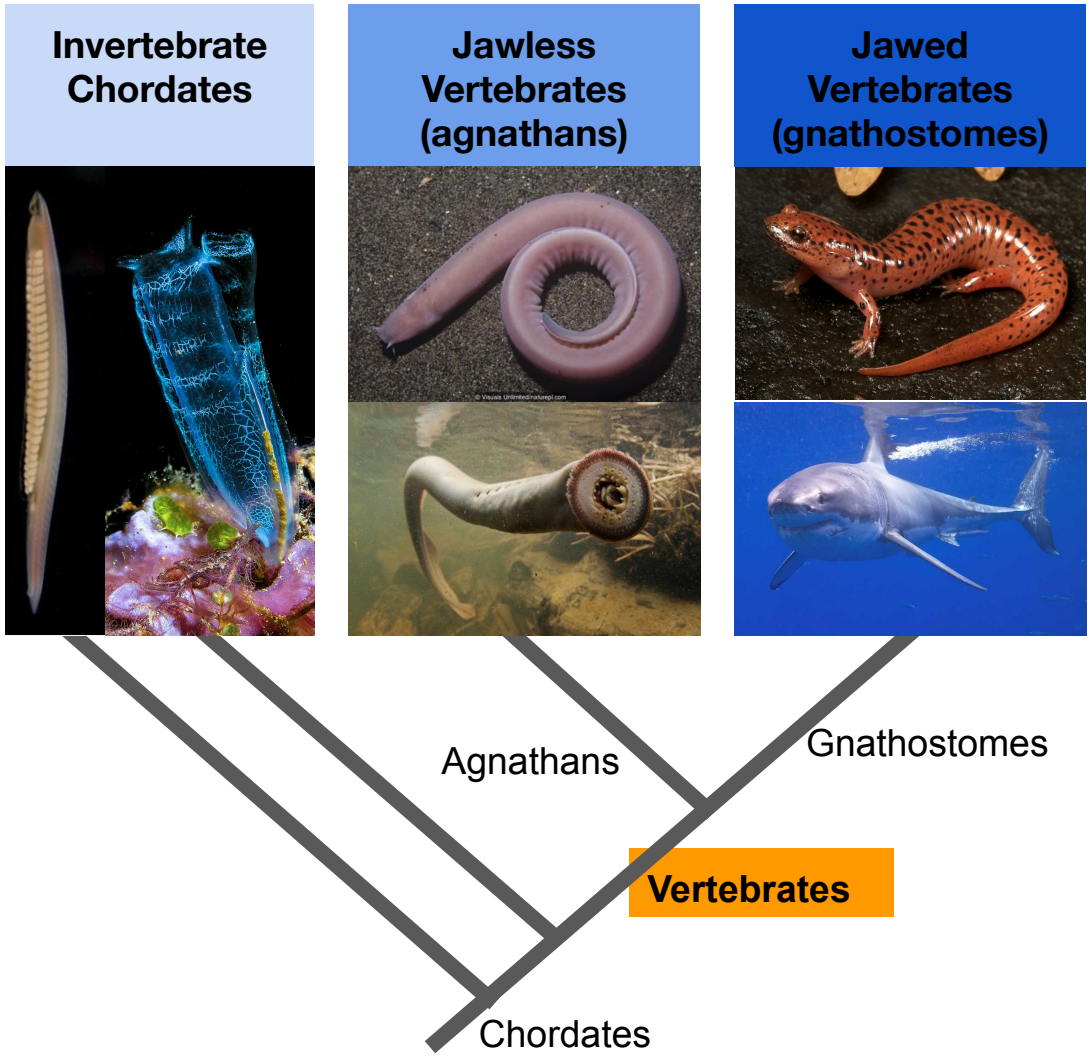


Chordates have:

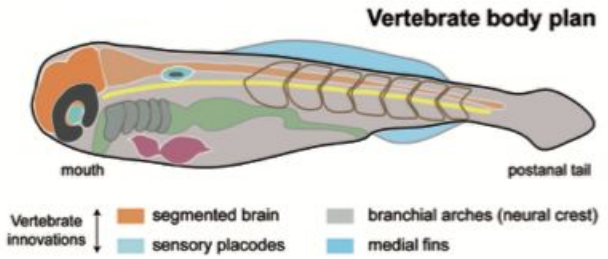
- **Notochord** (stiff rod along the back)
- **Dorsal nerve cord**
- **Pharyngeal slits** (for breathing)

These are **synapomorphies**: unique defining characteristics

Vertebrates



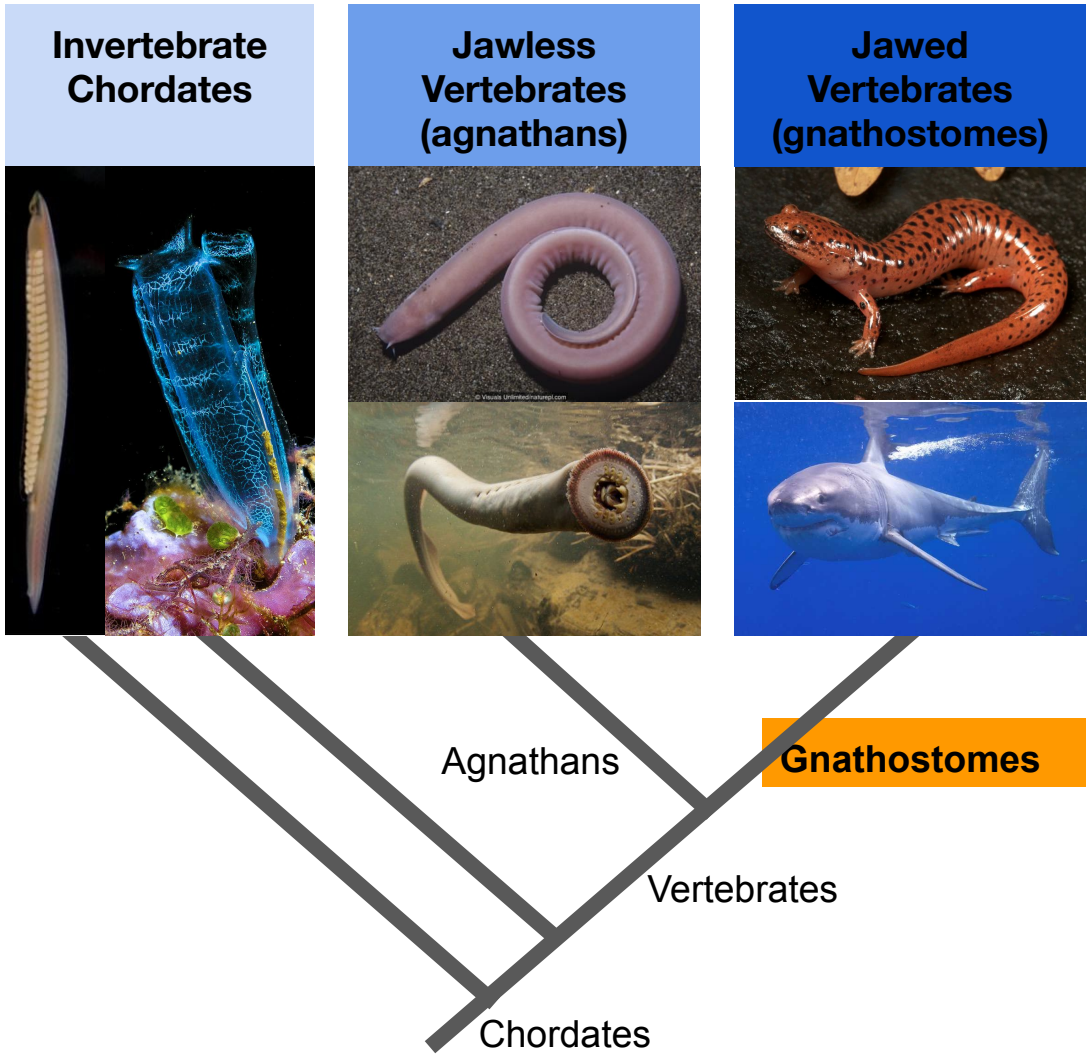
Vertebrates are in the phylum chordata



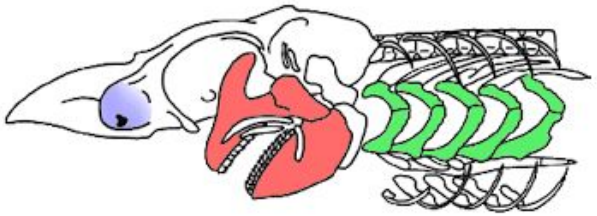
Vertebrates have all the chordate characters plus:

- Segmented **Brain**
- Medial **Fins**
- **Branchial arches** (which develop into gills and other things)
- Paired eyes, olfactory bulbs and ‘ears’

Vertebrates



Vertebrates are in the phylum chordata



Gnathostomes have all the chordate and vertebrate characters plus:

- **Jaws**
- **Teeth**
- Mineralized **Bone**
- **Paired appendages** (pectoral and pelvic i.e. fins or legs)

Vertebrate Clades

Jawless Fishes (Agnathans)



Lampreys

Blood sucking ‘vampire eels’ with keratinous ‘teeth’ and a ‘sucker’



Hagfish

Slimey, blind, bottom-dwelling ‘eels’ with rasping keratinous teeth

Jawed Fishes (Gnathostomata)



Sharks and rays: cartilaginous skeletons, boney teeth

Chondrichthyes: cartilaginous fishes



‘Classic’ fishes: boney skeleton and fins with rays

Actinops: ray finned fishes

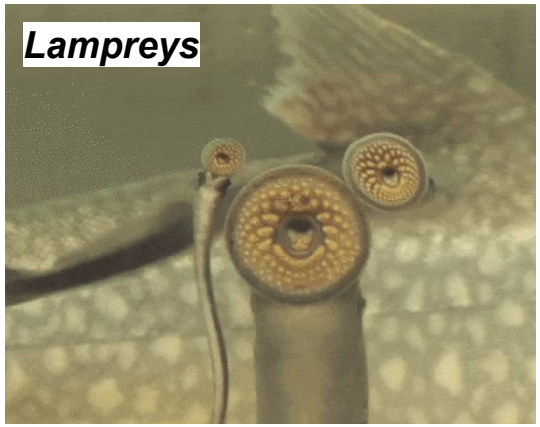


Fishes with fleshy lobed fins and ‘fishes’ with legs (tetrapods)

Sarcops: lobe finned ‘fishes’

Vertebrate Clades

Jawless Fishes (Agnathans)



Blood sucking 'vampire eels' with keratinous 'teeth' and a 'sucker'

Great Lakes Fishery Com.



Vancouver Aquarium

Jawed Fishes (Gnathostomata)



Sharks and rays: cartilaginous skeletons, boney teeth

Chondrichthyes: cartilaginous fishes



'Classic' fishes: boney skeleton and fins with rays

Actinops: ray finned fishes



Fishes with fleshy lobed fins and 'fishes' with legs (tetrapods)

Sarcops: lobe finned 'fishes'

Vertebrate Clades

Chordata



Invertebrate Chordates

Vertebrata



Jawless Fishes

Gnathostomata





Jawed fishes and tetrapods

Ocean Life
4.4 Vertebrates

Intended Learning Objective: Define vertebrates in terms of their constituent clades and characteristics

Now take a look at the associated task/question for this section

Robert Sansom, University of Manchester



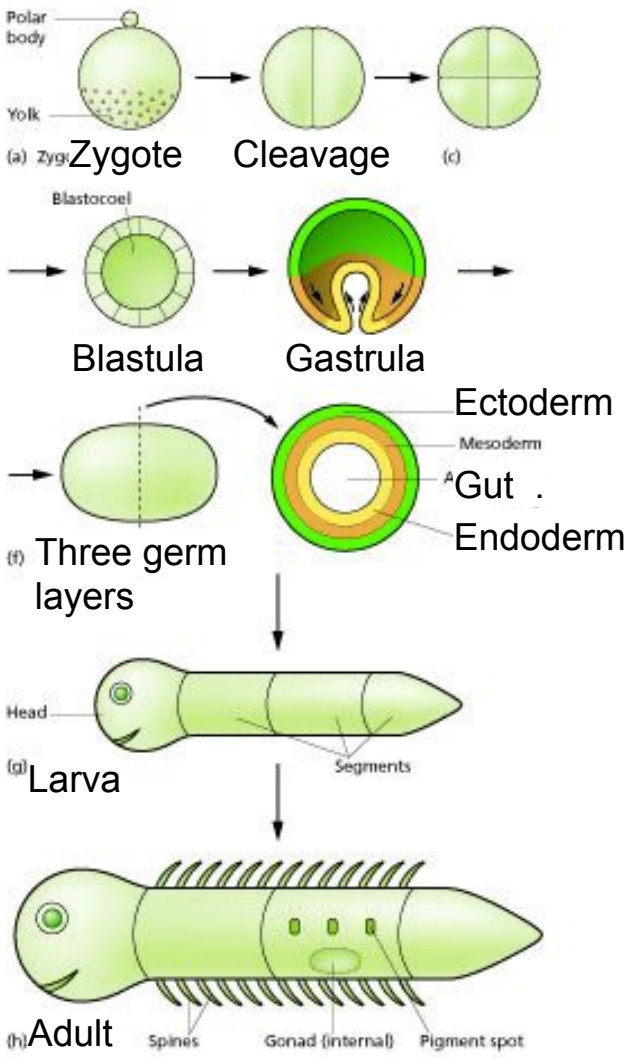
Part 4
Ocean Life

4.5 Developmental Biology

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Animal Development



Sperm and egg combine to make a **zygote**, which undergoes cleavage (cell division)

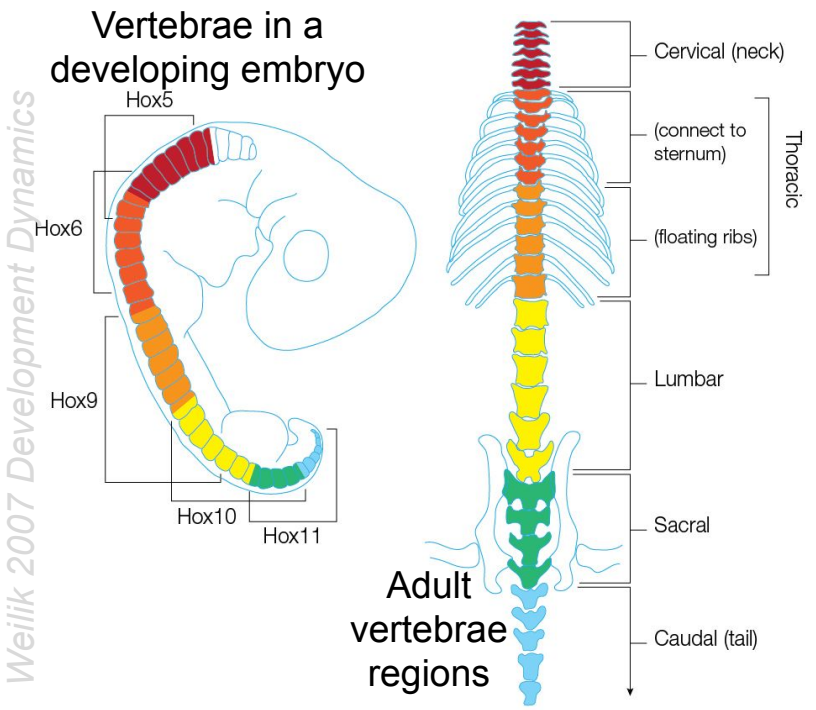
The first stage is a **Blastula** (ball of cells) and then **Gastrula** following invagination of the surface to create an internal gut cavity

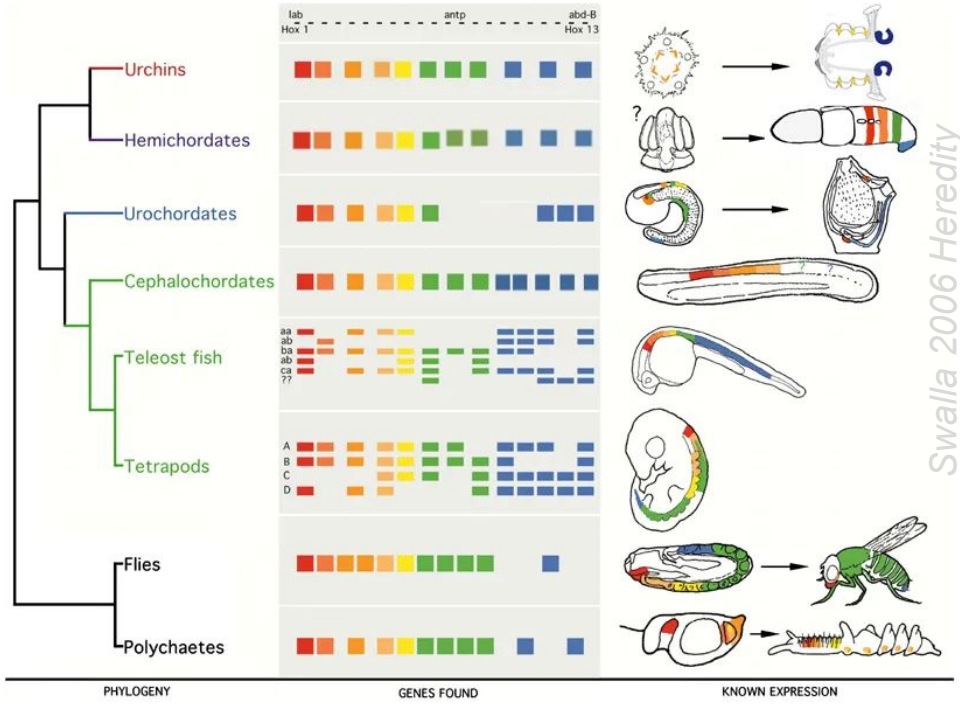
The developing body has germ layers (**endoderm** on the outside, **ectoderm** surrounding the gut), like a tube

This then develops into a larvae (for example specification of a head and body segments)

Finally the larvae becomes an adult, including gonads, appendages etc.

Gene Expression Patterns

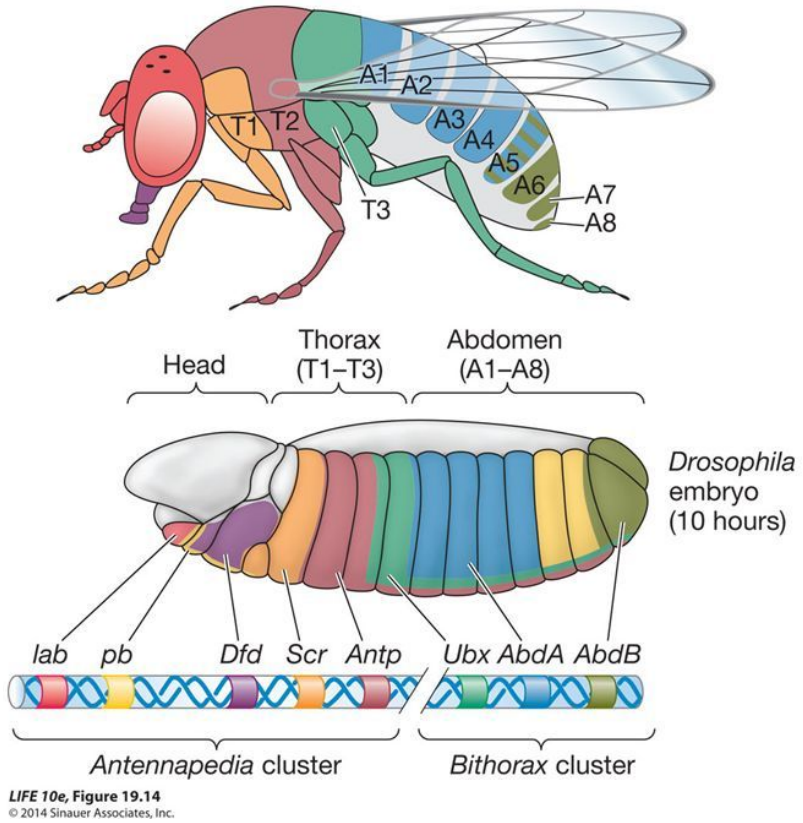




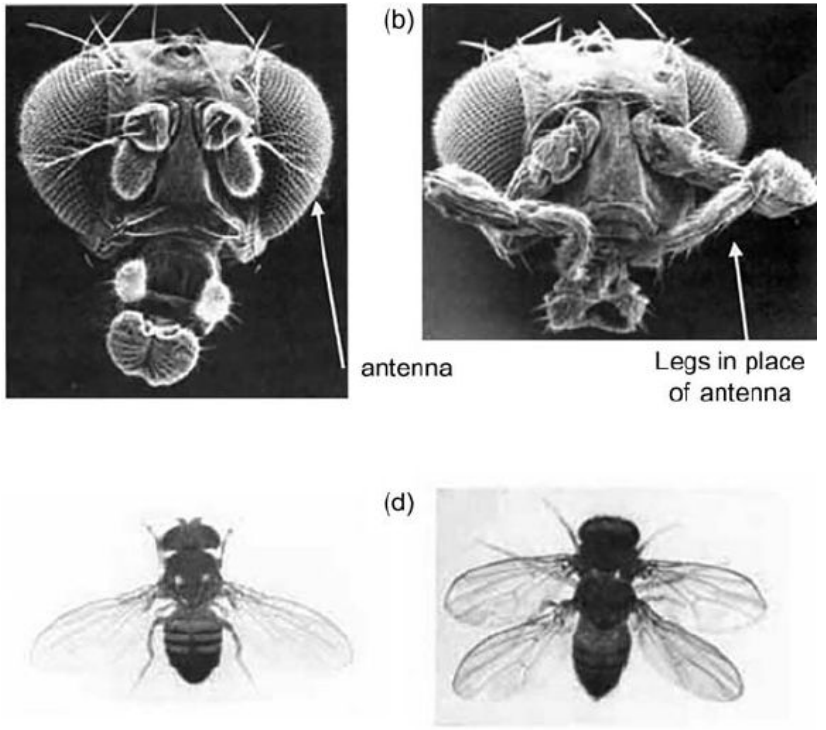
As the animals develop, **Homeobox genes** (Hox) are expressed in different regions of the body (e.g. segments)
This specifies the type of anatomy in that region

Hox genes are arranged on chromosomes and are highly conserved across animals groups
Interesting things happen in the vertebrates, as they have multiple copies of each Hox gene (gene duplication)

Hox Genes



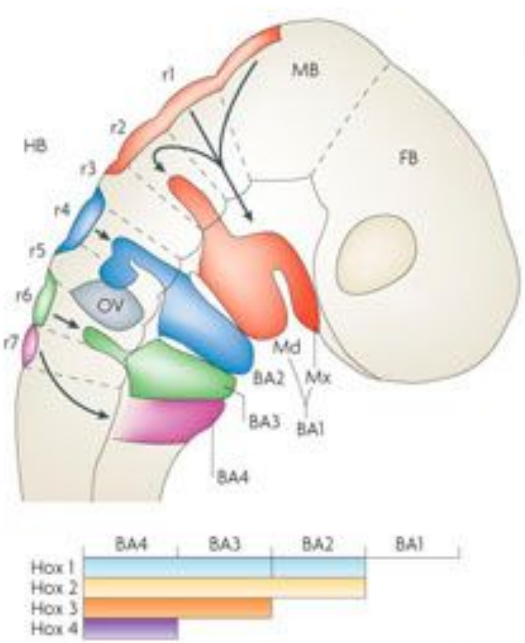
An example of segment specification in a developing fruit fly

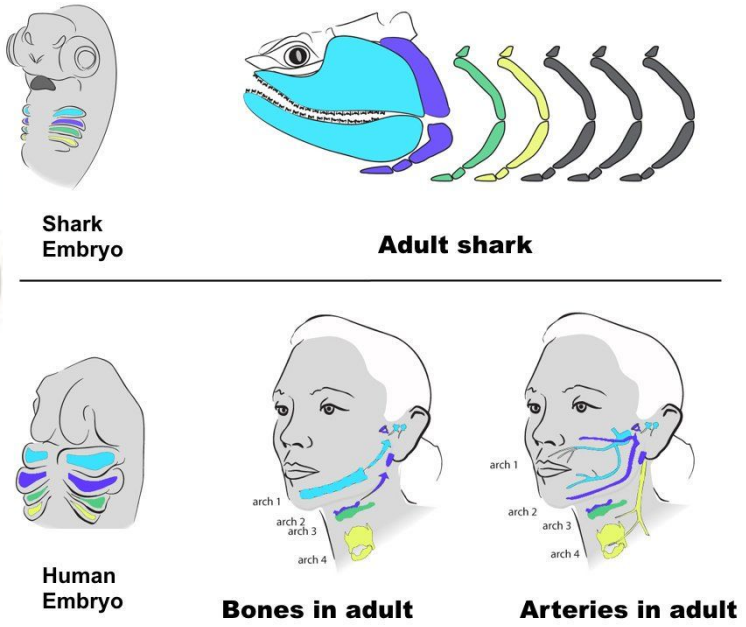


Allouba et al 2015 GCSP

Hox gene **mutants** in the fruit fly *Drosophilia* can have very different morphologies

Branchial Arches and Jaws

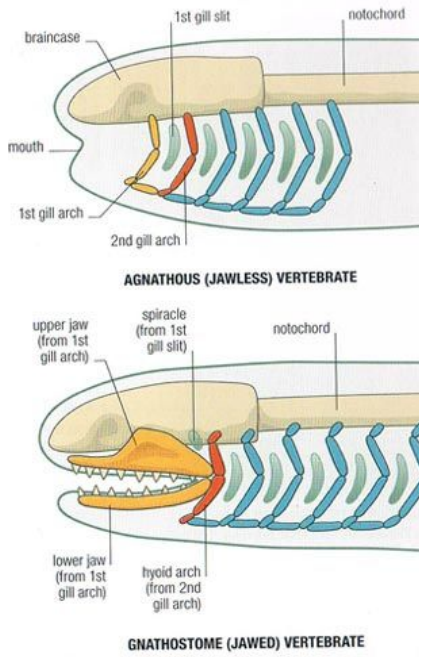




Guthrie 2007 Nat Rev Neuro

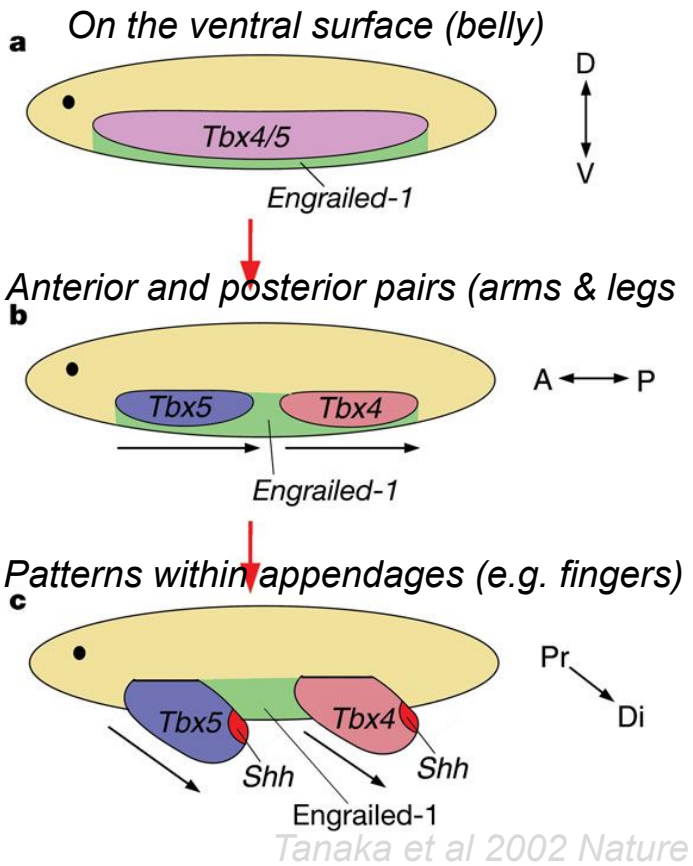
N Shubin/K Monoyios/Your Inner Fish

Vertebrates embryos show some segmentation
Embryonic expression patterns of Hox genes relate the fate of the **branchial arches** (BA)
For example:
No Hox expression = the mandible arch (BA1, the jaw)
Hox 1+ Hox 2, but not Hox 3 = the hyoid and facial (BA2)



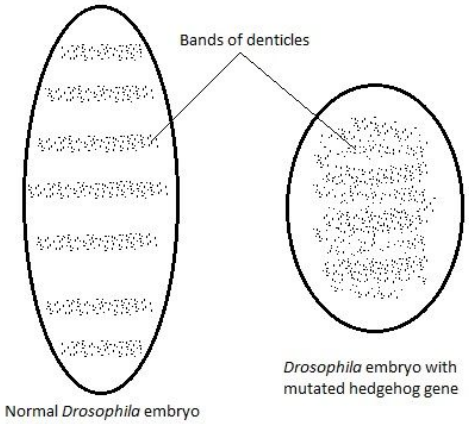
The vertebrate jaw might have evolved from the first branchial arch of a jawless ancestor

Development of Appendages (fins and limbs)



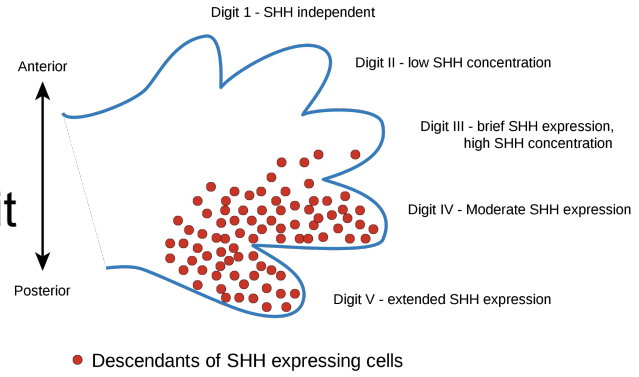
Gene expression patterns also determine where and how appendages develop

For example, the **Sonic Hedgehog** protein is a key component of development and specifies tissue fates in combination with others



Name for mutant observed in fruit fly that looked like a hedgehog

In vertebrate limbs, Sonic hedgehog specifies digit identity (e.g. fingers)



Ocean Life
4.5 Developmental Biology

Intended Learning Objective: Outline the basic principles of developmental biology in the context of vertebrate body plans

Now take a look at the associated task/question for this section

Robert Sansom, University of Manchester



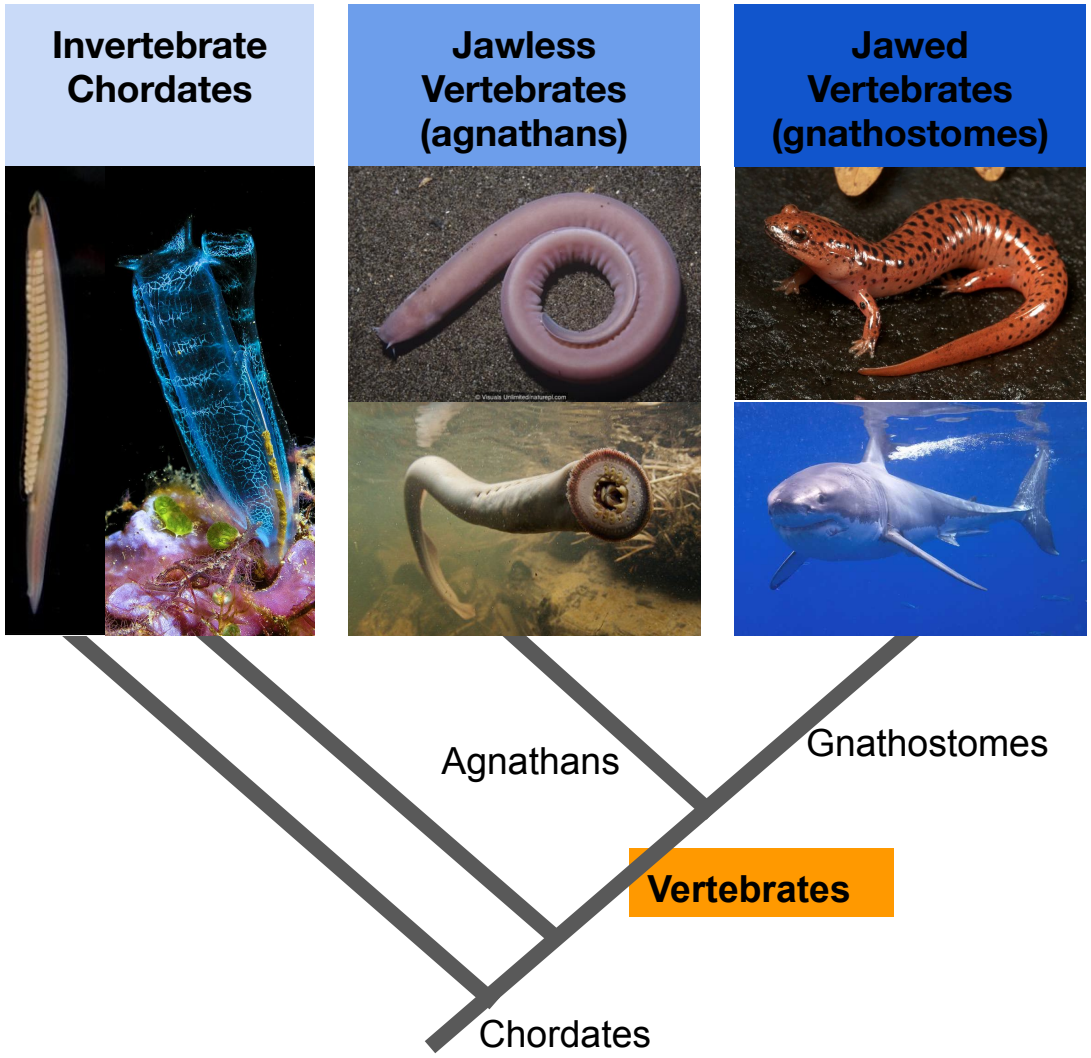
Part 4
Ocean Life

4.6 The Age of Fishes

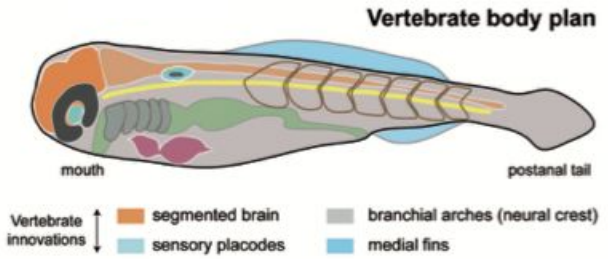
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University of Manchester



Vertebrates



Vertebrates are in the phylum chordata

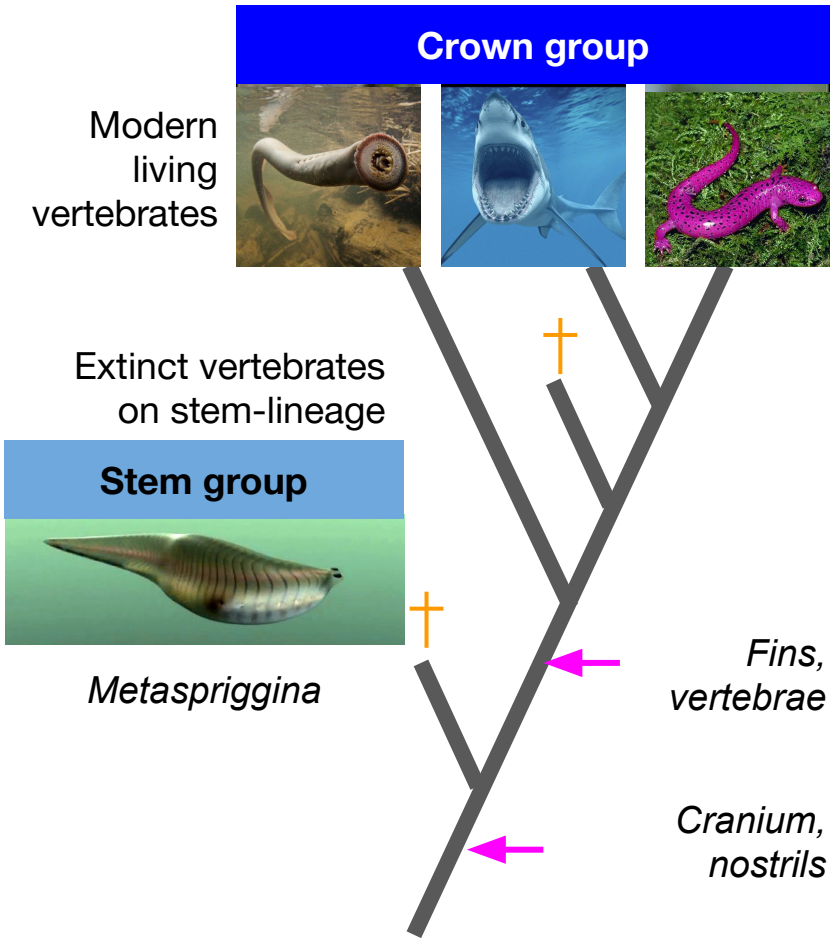


Vertebrates have all the chordate characters plus:

- Segmented **Brain**
- Medial **Fins**
- **Branchial arches** (which develop into gills and other things)
- Paired eyes, olfactory bulbs and ‘ears’

Cambrian Vertebrates

The vertebrate stem group



Metaspriggina

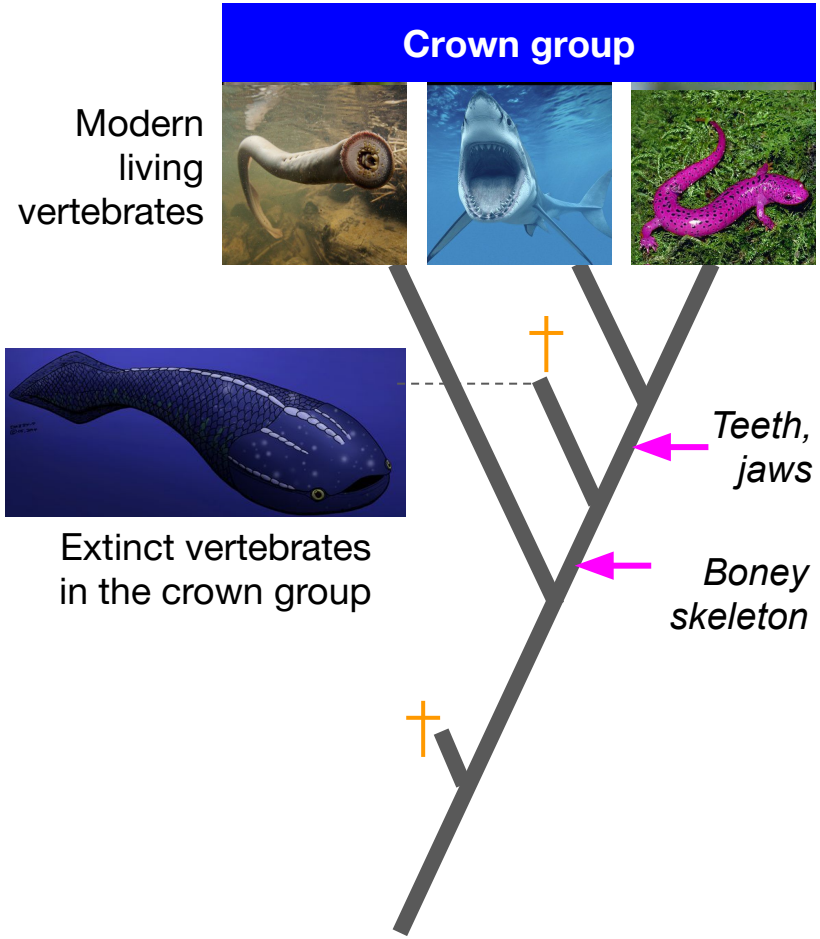


A Cambrian stem vertebrates from the Burgess Shale with paired eyes and branchial arches (gills)

Conway Morris & Caron 2014 Nature

Ordovician Vertebrates

The vertebrate crown group



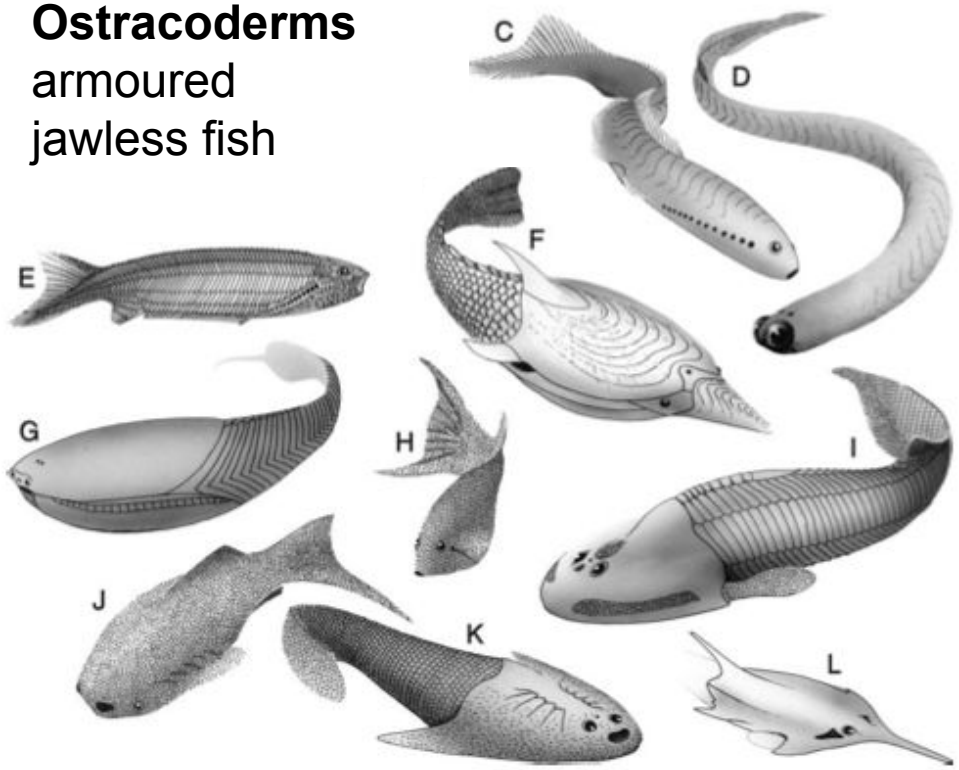
Sacabambaspis



An Ordovician jawless fish with a boney dermoskeleton and paired nostrils

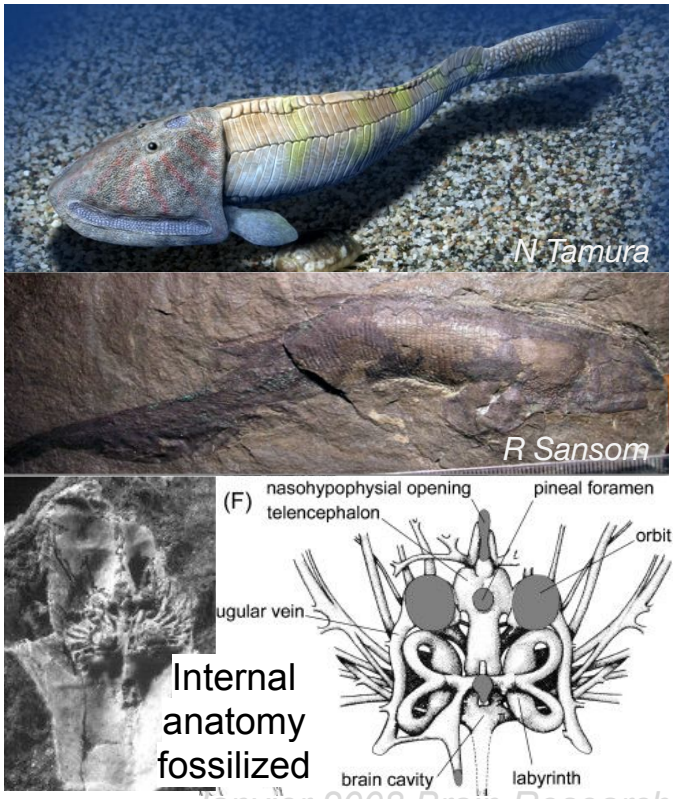
Silurian Vertebrates

Ostracoderms
armoured
jawless fish



Donoghue 2000 Biological Reviews

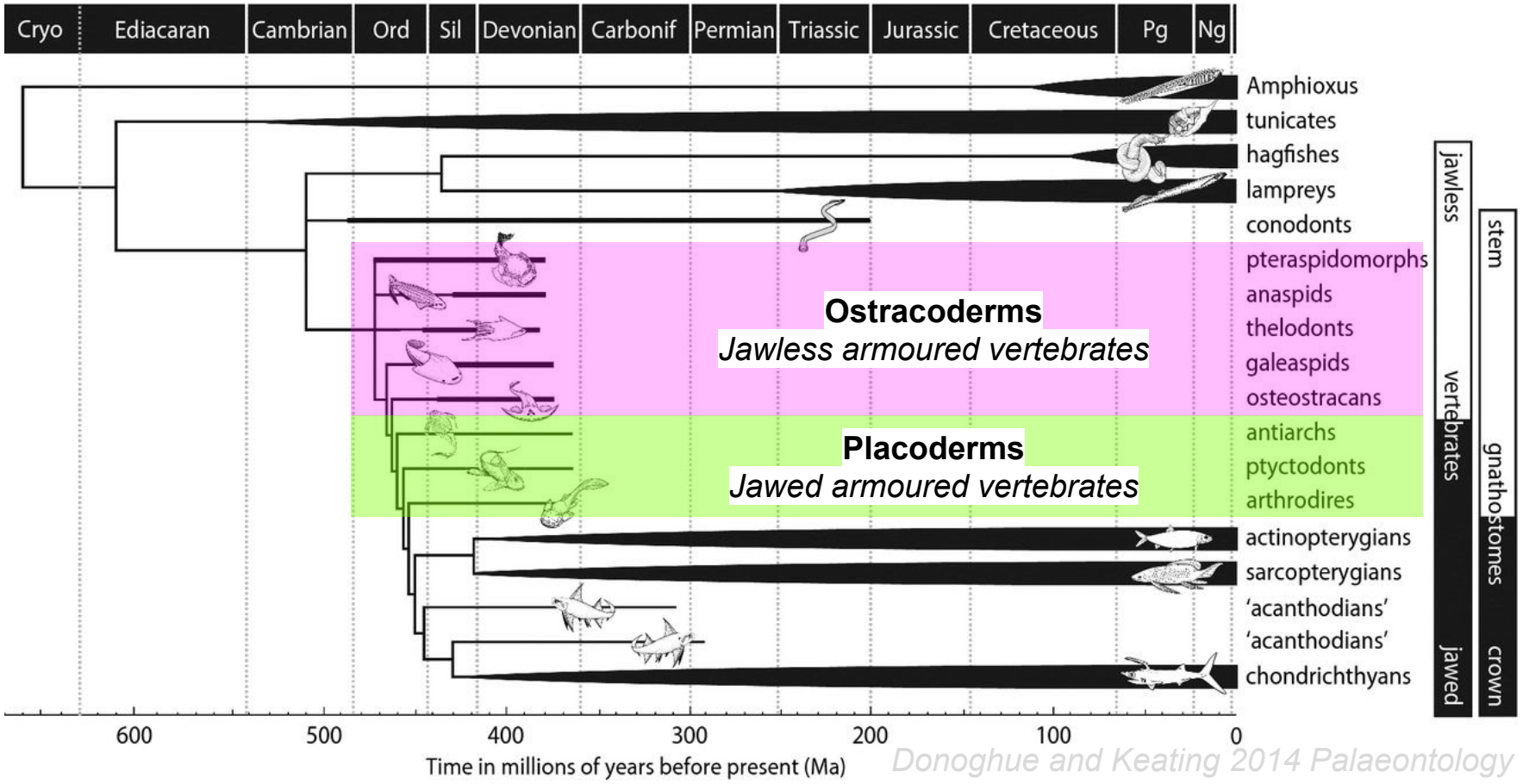
Sacabambaspis is an **ostracoderm** (armoured jawless fish)
They were the diverse dominant vertebrates in the Silurian



For example, ***Cephalaspis*** has paired fins but a lamprey like brain and single nostril opening


446 309 765 YEARS AGO
Discovery

Stem-gnathostomes

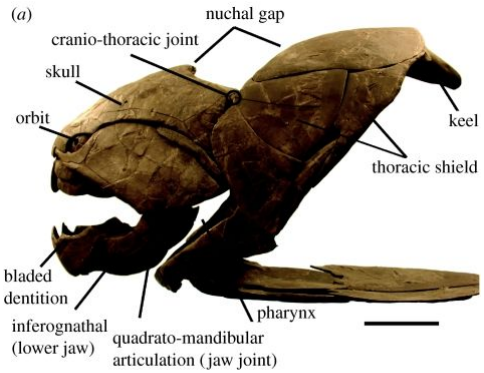


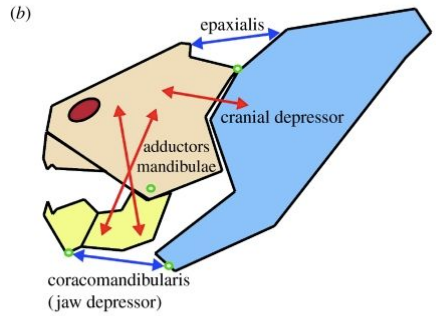
Ostracoderms are **stem-gnathostomes:** They reveal the stepwise assembly of jawed vertebrate characters along the gnathostome stem: boney skeleton, advanced brain, appendages, sensory systems, even teeth and jaws

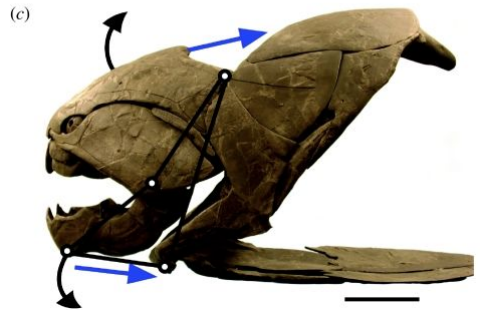
Devonian Vertebrates



Placoderms originated in the Silurian but diversified and dominated in the Devonian
They are armoured **jawed** fish, also stem-gnathostomes
They have biting jaws, primitive ‘teeth’, and pelvic appendages





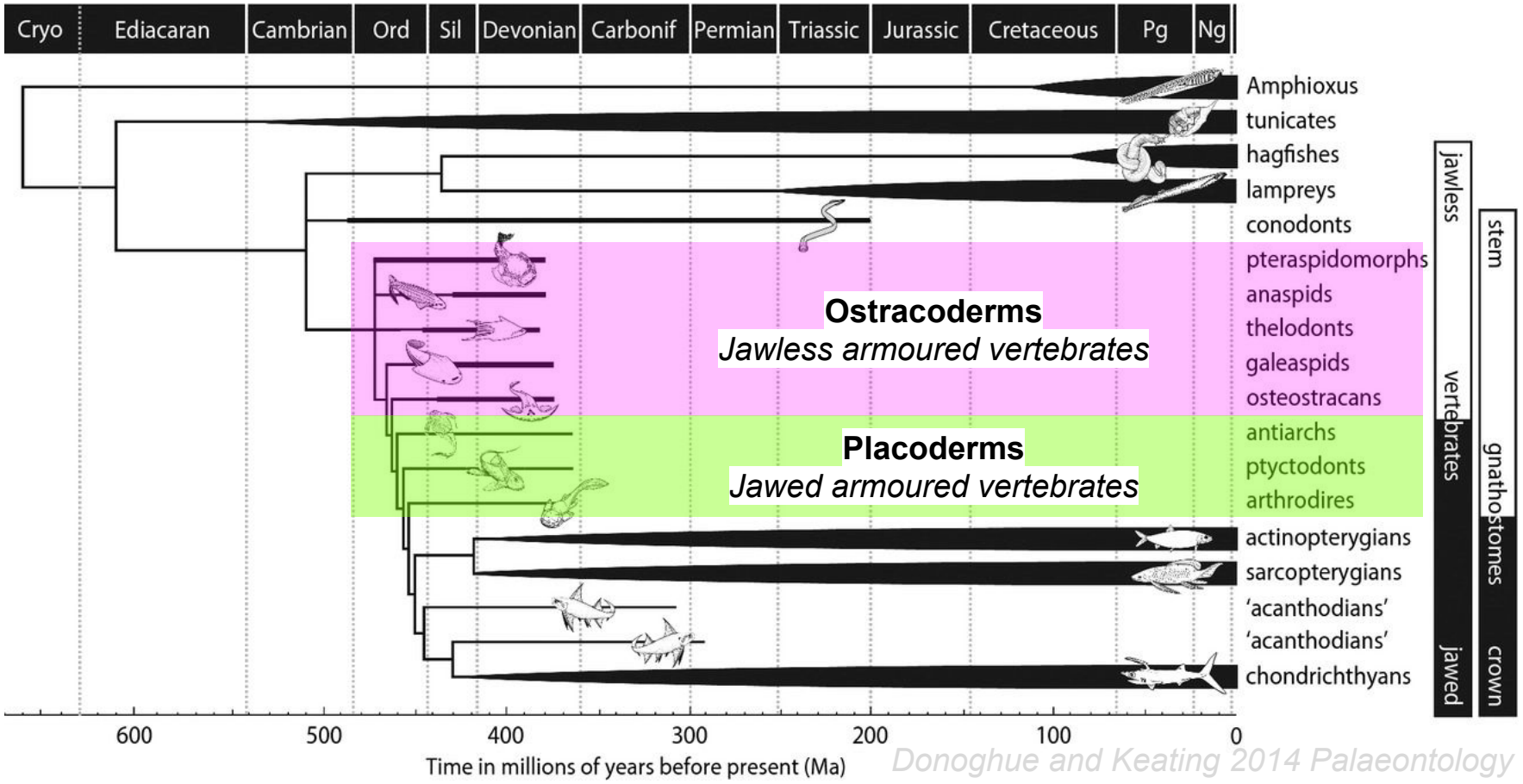


Placoderms includes the first ‘mega-predators’ e.g. ***Dunkleosteus*** (6m long)

The bladed jaws of *Dunkleosteus* had extra-ordinary biomechanical **biting strength**: 5000 Newtons (stronger than a great white shark)

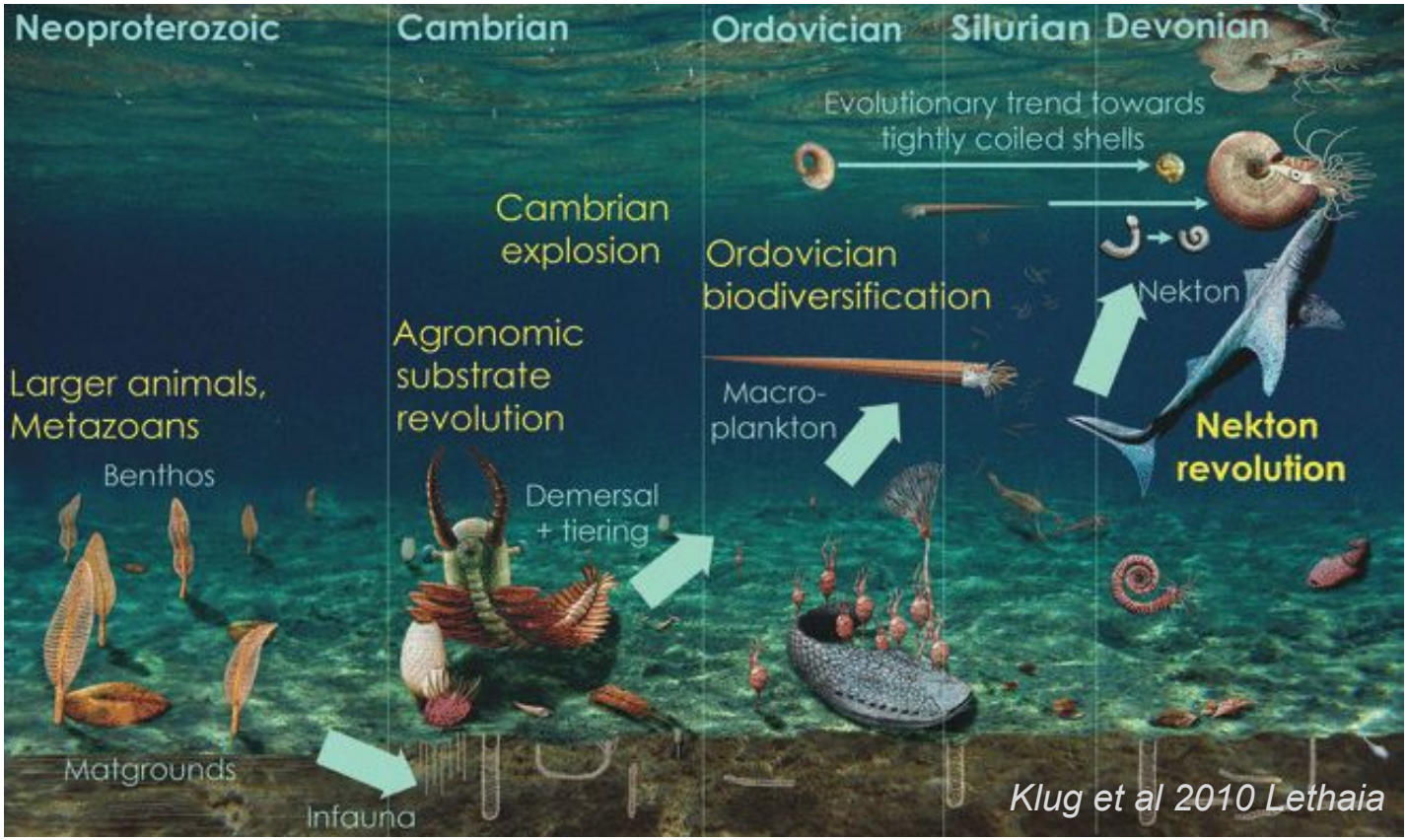
Anderson and Westneat 2007 Biology Letters

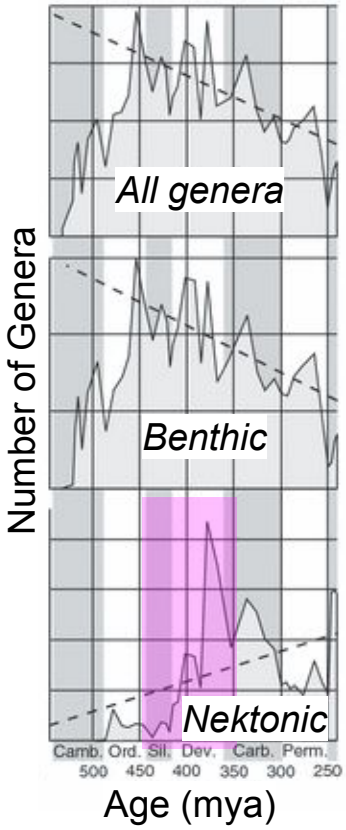
Stem-gnathostomes



Placoderms are also **stem-gnathostomes:** they have jaws, pectoral and pelvic appendages, and some sort of 'proto'-teeth, but they don't have the full suite of gnathostome characters

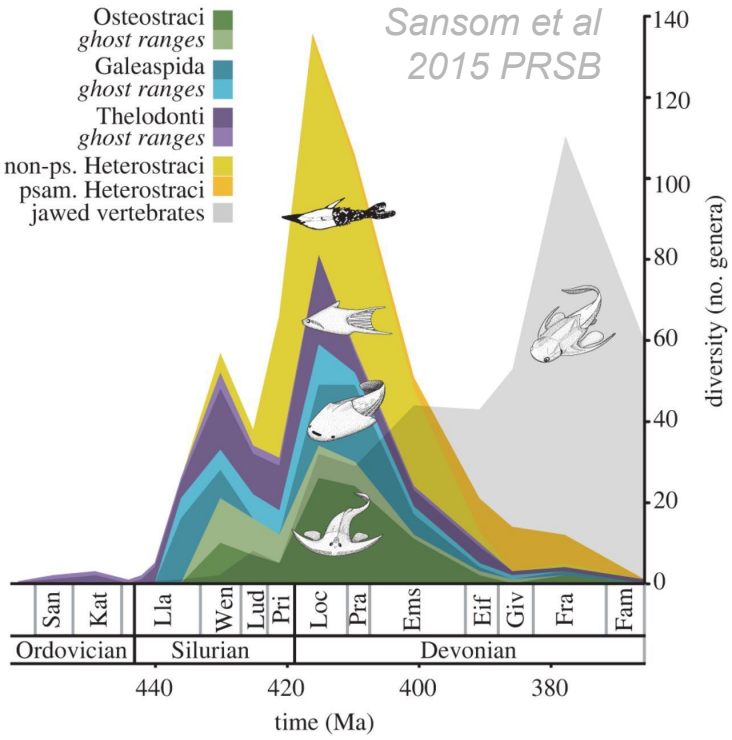
Ecological complexity



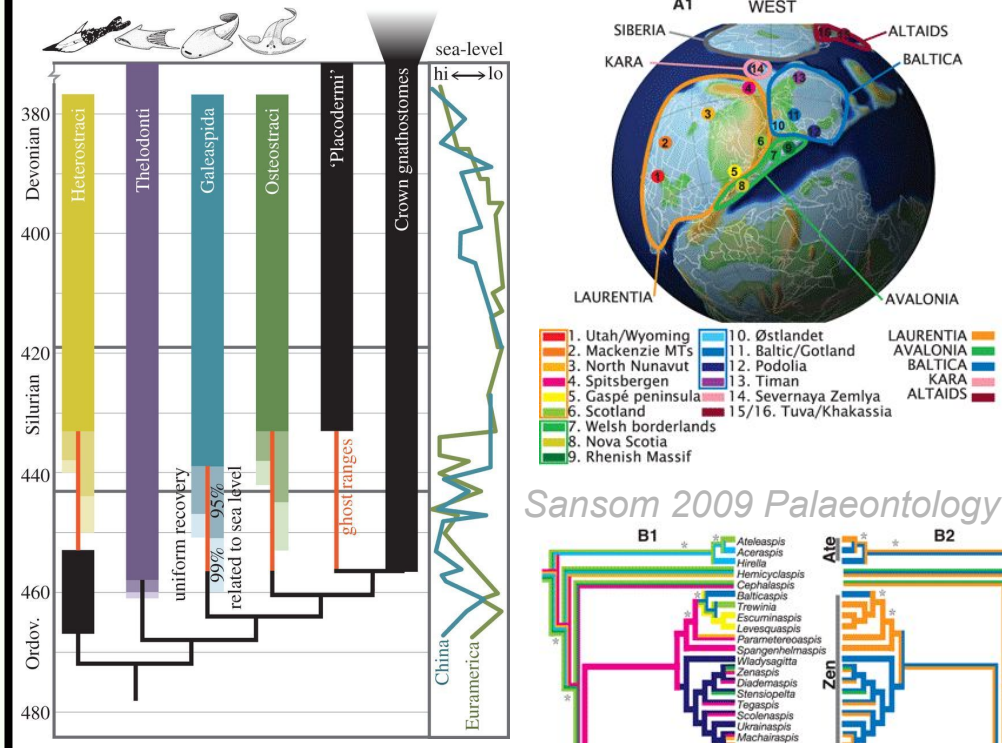


Large Devonian swimming predators is another advance in ecological complexity
From simple flat Ediacaran ecosystems, burrowing Cambrian ecosystems, planktonic Ordovician ecosystems, to the **Devonian Nekton revolution**: occupation of the deep water column

Devonian Decline



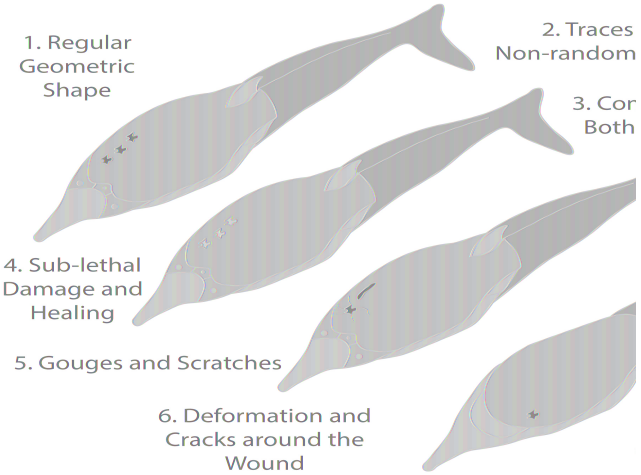
After their early peak, ostracoderms undergo a slow **Devonian decline & extinction**
Jawed vertebrates diversify later but also suffer at **late Devonian extinction events**



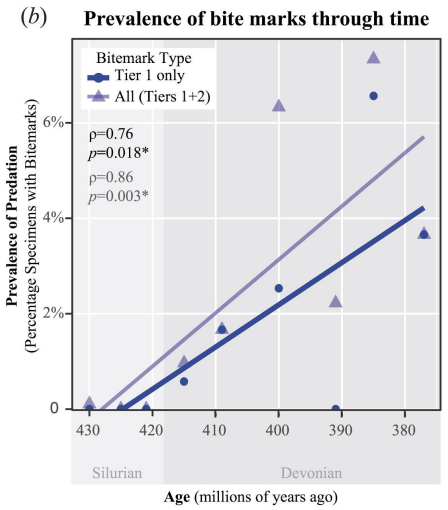
Collision of North America and Europe to form **Euramerica** continent
Ostracoderms were restricted to shallow water environments, little migration
They couldn't cope with rising sea level and changing conditions

Ostracoderms predated by Jawed Vertebrates

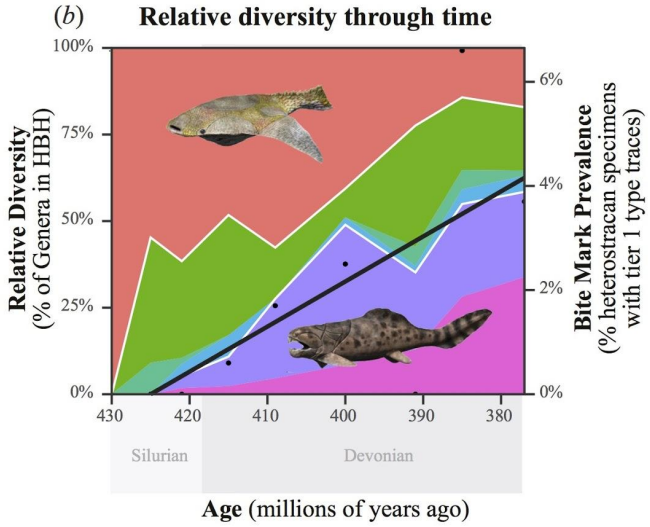


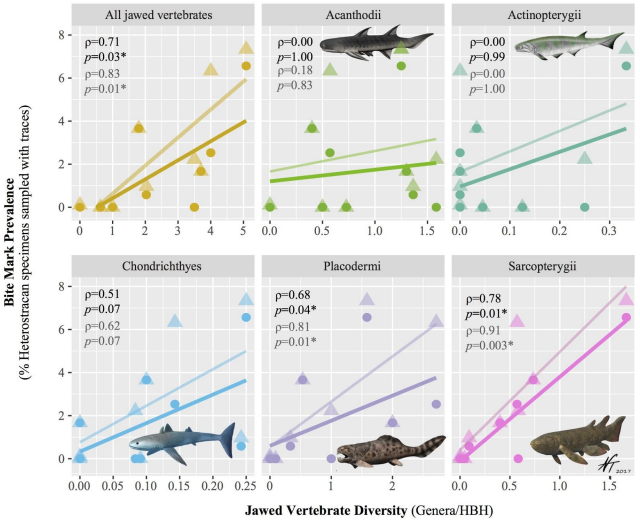


Randle and Sansom In Review



Ostracoderm fossils have **bite mark** traces which increase in prevalence through the Devonian

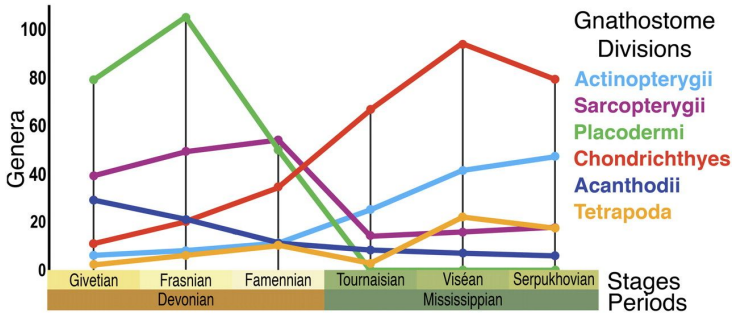




Bite marks are correlated with diversity and occurence of **jawed vertebrates**

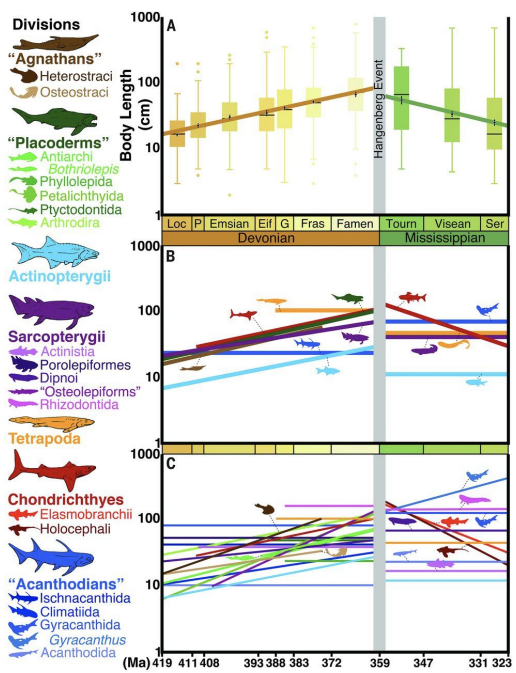
Ostracoderms might have been predated to extinction

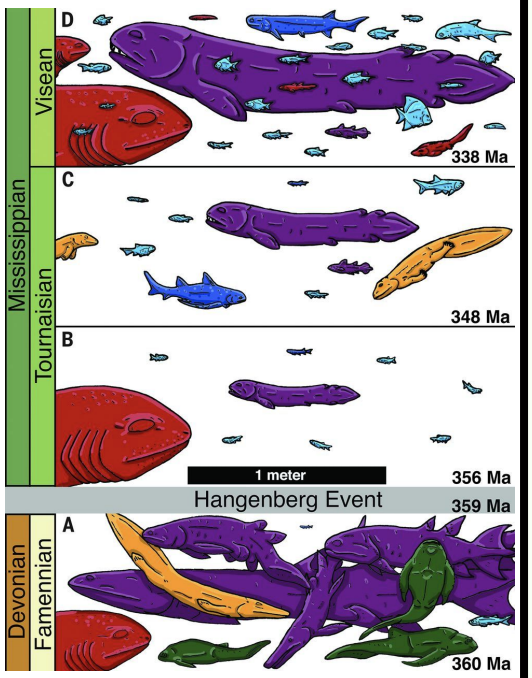
Late Devonian Extinction(s)



Within the **jawed vertebrates,** placoderms and lobe-finned fishes decline at End Devonian
More modern clades (Chondrichthyes and Actinopterygians) thrive after

The cause(s) of the end Devonian extinctions is unclear, but there were at least two pulses of extinction

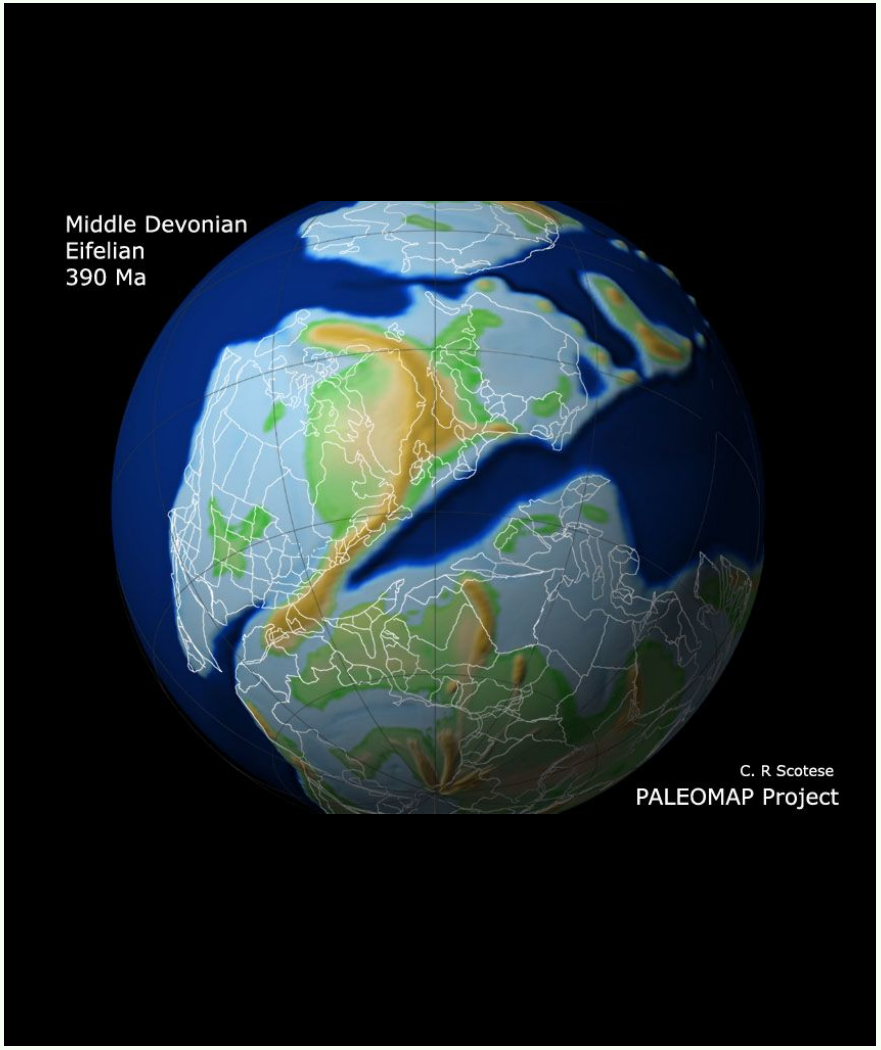




Body sizes was increasing in all vertebrates toward the end Devonian
The extinction event(s) wiped out large bodied vertebrates
Mostly small bodied afterwards with slow recovery to former sizes

Silurian and Devonian Summary

- Vertebrates originate in the Cambrian (*Metaspriggina* from Burgess Shale)
- Armoured jawless fishes (ostracoderms) diversify during Ordovician and Silurian
- Jawed vertebrates diversify and dominate in the Devonian, potentially predating and out-competing ostracoderms



Ocean Life
4.6 The Age of Fishes

Intended Learning Objective: Apply knowledge of the fossil record to interpret the origins of vertebrates and jawed vertebrates

Intended Learning Objective: Summarize the rise and decline of vertebrate clades during the Devonian in light of ecological factors

Robert Sansom, University of Manchester

